

Who Benefits from Benefits?

Municipal Consequences of Federal Cash Transfers

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May 21, 2026

Abstract

Cash transfer programs are centrally designed to limit local discretion. Yet in Brazil's *Bolsa Família*, eligibility rules and benefit values are managed federally while beneficiary registration is operated municipally through local welfare offices (*CRAS*). A 2009 federal reform revised the cross-municipal allocation formula, generating exogenous variation in authorized enrollment that we use to instrument realized expansion across the 2,105 municipalities (approximately 38 percent of the country) where the sitting mayor sought reelection in 2012. A one-standard-deviation increase in instrumented enrollment raises incumbent vote share by 2.94 percentage points, large enough to neutralize the typical inter-election incumbency disadvantage. The mechanism is gated at two binding margins: federal authorization translates weakly into realized expansion without a pre-existing *CRAS*, and approximately 93 percent of the within-*CRAS* identifying mass concentrates where the incoming mayor replaced the front-line manager between 2008 and 2009. Three quarters of those replacements bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles. The substitutes are not pre-selected on prior partisan affiliation but run for elected office at higher rates than stayers and align with the mayor's coalition, consistent with program expansion raising the visibility of the managerial post and with those individuals leveraging it to launch local electoral careers. Centralized design does not shield delivery from local politics.

Keywords: cash transfers; electoral returns; state capacity; *Bolsa Família*.

JEL Codes: D72, H53, H75, I38, O17.

I thank Solange Ledi Gonçalves and Valdemar Pinho Neto for invaluable guidance throughout this project. Diogo Souza, Jose Zobarán, and Marcos Lisboa for detailed feedback on earlier drafts. I am also grateful to Bruno Ferman and seminar participants at FEA-USP for helpful comments and suggestions. I thank the staff at *Ministério de Desenvolvimento Social* for access to the *Cadastro Único* administrative data. All remaining errors are my own.

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1 Introduction

Governments increasingly rely on cash transfer programs to alleviate poverty while limiting local political discretion. By standardizing eligibility rules, fixing benefit values by federal decree, and processing payments through centralized banking systems, these programs are designed to reduce the scope for patronage and discretionary allocation (Fiszbein and Schady, 2009). Yet access often depends on local bureaucracies controlled by elected officials. Whether control over program registration rather than program design is sufficient to generate electoral returns is the question we address.

We study this question in the context of Brazil's *Bolsa Família*, one of the largest conditional cash transfer programs in the world.¹ Although eligibility rules, benefit size, and payments are determined by the federal government, municipalities remain responsible for identifying and registering eligible households, mostly through local welfare centers, the CRAS (*Centro de Referência de Assistência Social*). This institutional arrangement creates a narrow but important margin of local discretion: while mayors cannot alter transfer amounts, redirect funds, or withhold payments, they control the administrative pipeline through which households may gain access to the program.

This margin operated through a registration process that required active municipal effort. Identifying potentially eligible families meant locating them, conducting household interviews, and collecting documentation; municipal authorities controlled the local registration apparatus, which gave them influence over the pace, intensity, and reach of enrollment efforts.² In practice, two municipalities facing similar levels of underlying need could generate very different realized coverage depending on how actively the local administration pursued registration.

Our empirical strategy exploits a 2009 reform that revised the formula used to compute municipal *Bolsa Família* quotas. Throughout this period, each municipality faced a federally determined quota, that is, a ceiling on the number of families that could be served under the program.³ The 2009 reform updated the formula used to calcu-

¹By June 2006, *Bolsa Família* covered 11.1 million families, approximately 46 million people, or about 25% of the Brazilian population. By June 2015, the program supported 13.8 million families, with an average monthly benefit of R\$167.15 and a budget equivalent to 0.5% of GDP.

²Hellmann (2015) describes enrollment as a municipal responsibility carried out in three stages: identification and localization of eligible families, household interviews, and digital entry of the information into the federal system. He reports that as of September 2014, 77% of registrations were made without a home visit. Municipalities were responsible for preparing the local structure for registration, training personnel, promoting awareness campaigns, and designing *active search* strategies to reach eligible families outside the existing registry.

³Lindert et al. (2007) explains that *Bolsa Família* maintained municipal program quotas based on local poverty estimates. Original municipal allocations were constructed by combining the 2001 PNAD with

late those municipal ceilings using census and survey information, generating large and heterogeneous changes in the number of families each municipality was authorized to serve.⁴ Crucially, the reform expanded the ceiling, not the registry: although already-registered eligible families were absorbed automatically, the magnitude of the expansion left a large residual margin of unfilled slots whose realization depended on local governments' effort to identify, locate, and register eligible families not yet in the federal database.

We combine administrative data on quotas, realized beneficiary counts, and local social-assistance infrastructure with municipal electoral data from the subsequent election.⁵ We use the federally induced quota shock as an instrument for realized beneficiary expansion and estimate its effect on incumbent electoral performance. The first stage is strong: municipalities receiving larger quota shocks expanded coverage substantially more than those receiving smaller shocks. This is the empirical counterpart to the institutional fact that federal authorization required local action to materialize.

Our findings trace the reform along four stages: the first-stage relationship between federal authorization and realized enrollment; the intention-to-treat effect of the authorization shock on electoral outcomes; the average causal response of electoral outcomes to realized enrollment that combines the two; and the institutional conditions under which these effects emerge.

The first stage confirms that federal authorization traveled to realized enrollment through the channel the institutional design contemplates. A one-standard-deviation increase in the quota shock, roughly 1,447 additional authorized slots, expanded realized beneficiary coverage by 0.21 standard deviations, approximately 1,015 additional families. Having established that the authorization shock moves realized enrollment, we ask what the reform itself delivers electorally, that is, the effect of expanding the federal ceiling, before separating out how much of that ceiling municipalities chose to fill. This intention-to-treat effect is 0.60 percentage points on the incumbent's vote share: the electoral consequence of authorizing more beneficiaries, integrating over the local take-up response documented in the first stage.

Scaling the intention-to-treat effect by the first stage gives the average causal re-

the 2000 Census, generating an initial national target of 11.2 million families when the program was created in 2003.

⁴Hellmann (2015) notes that *Bolsa Família* administered municipal quotas using estimates of poor families developed in partnership with IBGE and reports that, in 2009, the federal government refined the measurement of the target population by incorporating methodologies to capture income volatility among vulnerable households. These studies formed a major expansion of coverage, incorporating more than 1.6 million additional families.

⁵The analysis focuses on the 2008 and 2012 municipal elections and covers 2,105 municipalities.

sponse of electoral outcomes to realized beneficiary expansion. The 0.60 percentage points reported above correspond to the 1,015 additional families that each 1,447-slot authorization shock translated into on average; equivalently, an additional 1,000 realized beneficiaries raise the incumbent's vote share by approximately 0.59 percentage points. In standardized terms, a one-standard-deviation increase in instrumented enrollment, about 4,949 additional families, raises vote share by 2.94 percentage points. The estimate is robust to alternative instrument thresholds, different measurement windows for the endogenous variable, other outcome definitions, distinct weighting schemes, and inverse-probability weighting for selection into the rerunning sample.

These returns travel through the institutional layer that converts authorization into delivery. In the 690 municipalities without a pre-existing CRAS in 2008, the quota shock generates a first stage roughly a factor of three weaker than in municipalities with a CRAS, and the electoral effect cannot be identified at conventional precision for the no-CRAS subgroup. The Frisch-Waugh-Lovell decomposition concentrates approximately nine-tenths of the headline identifying mass on the 1,415 CRAS municipalities, where the implementation chain operates.

Within CRAS municipalities, a second binding margin operates through manager turnover. The within-CRAS first stage attenuates sharply where the *de facto* front-line manager who runs the CRAS unit day-to-day did not change between the 2008 and 2009 waves of the survey. Identification weights concentrate approximately 93 percent of the within-CRAS identifying mass on the manager-turnover subsample, and we report cell-by-cell Anderson-Rubin confidence sets confirming that within the no-turnover CRAS cell, the IV coefficient is not identified at conventional precision. The Wald test of equality between turnover and no-turnover first-stage coefficients rejects, and three quarters of the manager changes captured by our definition bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles.

The federal authorization causes substitution at incumbent CRAS units without causing new unit openings; the substitutes the shock attracts treat the post differently from inherited managers. Substitutes are not pre-selected on prior partisan affiliation, but they run for elected office at higher rates than stayers, mostly for city council, and they file new affiliations with parties in the mayor's electoral coalition rather than joining the mayor's party.

These findings place the paper in conversation with four related literatures. We speak first to the political economy of cash transfer programs. A large body of work shows that incumbents benefit electorally from the expansion of social transfers ([Manacorda et al., 2011](#); [De La O, 2013](#)). Much of this literature studies settings in which

local discretion is deliberately minimized, relying on centrally determined thresholds, randomized assignment, or other exogenous variation to identify electoral effects. We complement this work by studying a setting in which the program architecture is centralized but the last mile of delivery is not. Our findings suggest that centralizing transfer rules and payment flows is not sufficient to depoliticize redistribution when access to the program depends on a locally controlled registration bureaucracy. Even when mayors cannot alter eligibility criteria or benefit levels, control over the administrative pipeline through which households gain program access can still generate measurable electoral returns.

Our paper is especially closely related to [Frey \(2019\)](#), but the two designs identify electoral returns under sharply different degrees of mayoral involvement in the implementation chain. Frey exploits a discontinuity in federal funding for the *Family Health Program*, a primary-care program in which teams (a doctor, a nurse, an assistant nurse, and community health agents) follow a federally specified protocol of routine home visits to poor households. His mechanism runs through an information channel embedded in those visits: as part of the protocol, the teams disseminate *Bolsa Família* eligibility information, and new enrollment follows mechanically, without any discretionary action by the municipal executive. We exploit the complementary margin. In our setting, converting expanded federal authorization into realized enrollment requires active effort from the municipal welfare staff that the mayor hires, deploys, and directs to identify, locate, and register eligible families; the channel is the local administration itself. The two designs also identify on largely disjoint subsamples: Frey’s variation concentrates in small, low-development municipalities crossing the funding threshold; ours identifies on the complementary set with room to expand under the 2009 quota reform. Within his sample, coverage growth driven by an outreach channel that does not require mayoral effort reduces incumbency advantage and shifts the composition of challengers toward less clientelistic parties. Within ours, coverage growth driven by a channel that requires mayoral effort produces positive returns for the incumbent.

Second, we contribute to the literature on political accountability under shared governance. A central question in this literature is whether voters can correctly attribute responsibility when multiple levels of government jointly produce policy outcomes ([Przeworski et al., 1999](#); [Besley, 2006](#)). [Ferraz and Finan \(2008\)](#) show that voters punish corrupt mayors when randomly assigned audits reveal misuse of federal funds, especially where local media facilitate information transmission. [Avis et al. \(2018\)](#) show that audits also reduce corruption through a legal disciplining channel. Our setting differs because the relevant information concerns positive delivery rather than malfea-

sance. Voters observe the expansion of a valued program and update their beliefs about the incumbent's competence and effort. The mechanism we document is therefore closer to credit claiming than to sanctioning, but it operates through the same logic: voters use observable policy outcomes to infer politician type. In that sense, our results speak to a broader literature on attribution in multi-level settings by showing that voters can reward local incumbents even when the policy is federally funded and formally centralized, so long as local effort remains salient in implementation.

Third, we contribute to the literature on state capacity and the bureaucratic foundations of policy delivery. [Besley and Persson \(2009, 2011\)](#) establish state capacity as a precondition for credible policy delivery, and [Finan et al. \(2017\)](#) survey the personnel and bureaucratic infrastructure through which public programs are implemented. Our findings speak to a basic margin within this framework: the *existence* of local administrative infrastructure. Municipalities without a pre-existing CRAS converted essentially none of their quota shock into additional beneficiaries; municipalities with a CRAS translated the shock into expansion at a material rate. The electoral consequences inherit the presence threshold: the political returns to federal policy accrue only where local capacity exists to channel them.

Fourth, conditional on the presence of a delivery apparatus, our within-CRAS findings on bureaucratic turnover speak to the literature on the selection and politicization of frontline bureaucrats. The 2009 federal authorization shock causes turnover of the *de facto* CRAS manager, and the substitutes it attracts are not pre-selected on prior partisan affiliation. Two complementary mechanisms rationalize the pattern. First, program expansion raises the *de facto* attractiveness of the managerial post and pulls a higher-ability pool of candidates into it, consistent with [Dal Bó et al. \(2013\)](#). Second, once in the post, those individuals run for elected office at higher rates than stayers and align with the mayor's electoral coalition rather than switching to the mayor's exact party; this echoes [Colonnelli et al. \(2020\)](#), in which public-sector positions serve as channels into electoral careers, though our finding is sign-reversed since substitutes are not pre-selected on partisan ties. The pattern thus departs from the loyalist-allocation equilibrium in [Robinson and Verdier \(2013\)](#) and from the hiring-discretion turnover in [Akhtari et al. \(2022\)](#): here a federal program shock drives the turnover, and politicization happens downstream rather than at hiring. [Frey and Santarrosa \(2025\)](#) corroborate this reading: *Bolsa Família* interviewers who remain in the post align politically with the mayor while serving, with the effect concentrated where program exposure is highest. The position itself becomes politicized as the program expands; individuals are drawn into the post by the visibility the program creates, and once there they end

up in politics.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting of *Bolsa Família*, the quota system, the 2009 reform, and the data sources. Section 3 develops the conceptual framework: an augmented model of the federal-to-local implementation chain that isolates two binding margins for delivery, the presence of local welfare infrastructure and the alignment of front-line bureaucrats, and predicts that the electoral return to program expansion concentrates where both are active. Section 4 presents the empirical strategy, defines the instrument, and discusses the identifying assumptions. Section 5 reports the main results, the robustness checks, and the heterogeneity decomposition. Section 6 tests the framework’s mechanism predictions against the data and documents the chain of evidence on the substitute managers the shock attracts. Section 7 concludes.

2 Background and Data

This section describes the *Bolsa Família* program, focusing on institutional aspects essential for understanding its local implementation dynamics and political implications. The first subsection lays out the program’s design, the registration process, the quota mechanism, and the 2009 federal reform that generates our identifying variation. The second introduces the four data sources we combine, and the third reports the descriptive statistics used throughout the paper.

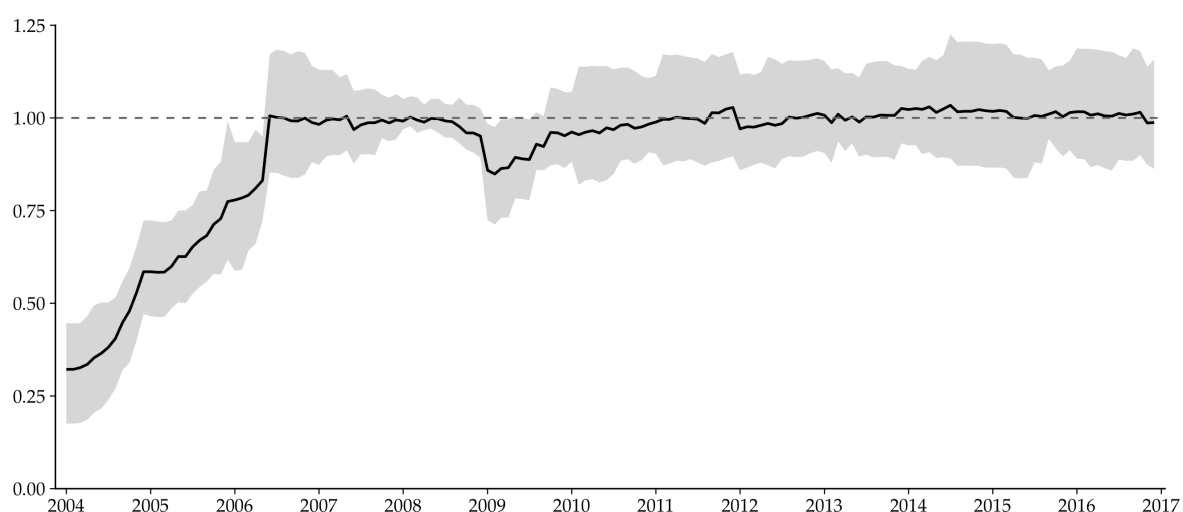
2.1 Institutional Background

Launched in 2004, *Bolsa Família* consolidated several pre-existing social assistance programs into a single conditional cash transfer system, becoming Brazil’s flagship poverty alleviation policy. The program provides monthly transfers to families below two income thresholds. Households with per capita income at or below R\$68 per month were classified as extremely poor and received an unconditional basic benefit of R\$68, plus conditional variable payments of R\$22 per child under 16 (up to three) and R\$33 per adolescent aged 16–17 (up to two), conditional on school attendance and routine health checkups. Families with per capita income between R\$68 and R\$137 qualified only for the child- and adolescent-linked variable benefits. The maximum monthly transfer per family was R\$200 and the average benefit stood at approximately R\$95. By September 2009, the program reached approximately 12.4 million families, or roughly a quarter of the Brazilian population.

Central to the program's operations is the *Cadastro Único* (*CadÚnico*), a national registry of low-income households that serves as the gateway to *Bolsa Família* and dozens of other social policies. The registry targets families with per capita income below half the minimum wage (R\$255 in 2010), a threshold substantially higher than the program's own eligibility lines, so that the pool of registered families considerably exceeds the set of actual beneficiaries.

Eligibility criteria are nationally uniform, benefit values are fixed by federal decree, and payments are processed and deposited directly into beneficiaries' accounts by the state-owned bank *Caixa Econômica Federal*. Mayors cannot alter transfer amounts, redirect funds, or withhold payments. What municipal governments control is the registration process itself.

Figure 1: Realized Enrollment as a Share of Authorized Quota



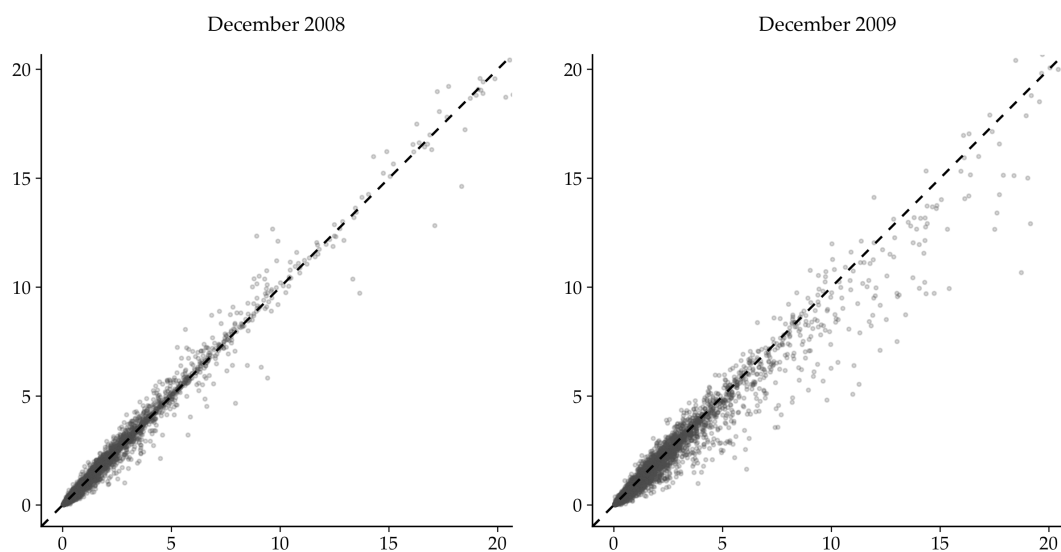
Note: Each point on the line is the cross-municipal mean of the ratio of realized *Bolsa Família* beneficiaries to the authorized quota in a given month, with each municipality weighted by its monthly authorized quota. The shaded band spans the weighted 25th to 75th percentile. The dashed horizontal line marks full take-up (ratio = 1). Before 2007, municipalities operated well below their authorized ceilings; the sharp convergence toward 1 reflects the rapid administrative build-up of the program in its first years. After 2007, the ratio fluctuates around 1, with a pronounced dip in mid-2009 as the reform's expanded quotas outpaced realized enrollment, the identifying variation we exploit.

Registration in the *Cadastro Único* requires a face-to-face interview: a social worker fills out a standardized questionnaire covering household composition, housing conditions, assets, employment, and income. In urban areas, families typically present themselves at a *Centro de Referência de Assistência Social* (CRAS). In rural and remote areas, however, eligible families cannot realistically be expected to travel to a registration point. There, coverage depends on active outreach: municipal teams must physically locate potential beneficiaries, travel to communities with limited road access, no tele-

phone coverage, and no internet, and conduct household interviews on site. Federal regulations encourage this practice by recommending that interviews take place at the family's home, both to facilitate verification and to reach populations that would otherwise remain invisible to the registry. In 2009, there were approximately 5,800 CRAS units distributed across roughly 4,000 of Brazil's 5,568 municipalities, and all data collection was done on paper.

Eligibility alone does not guarantee benefit receipt. Access to *Bolsa Família* depends on the availability of municipal quotas: caps on the number of beneficiaries per municipality, defined nationally and revised approximately every three years based on poverty estimates derived from census and survey data. These quotas function as official upper bounds on municipal enrollment, but the constraint is not always immediately binding. In practice, municipalities frequently operate near their ceiling, oscillating modestly above or below it from month to month before the federal priority mechanism reallocates unused slots.

Figure 2: Authorized Quota vs. Realized Enrollment

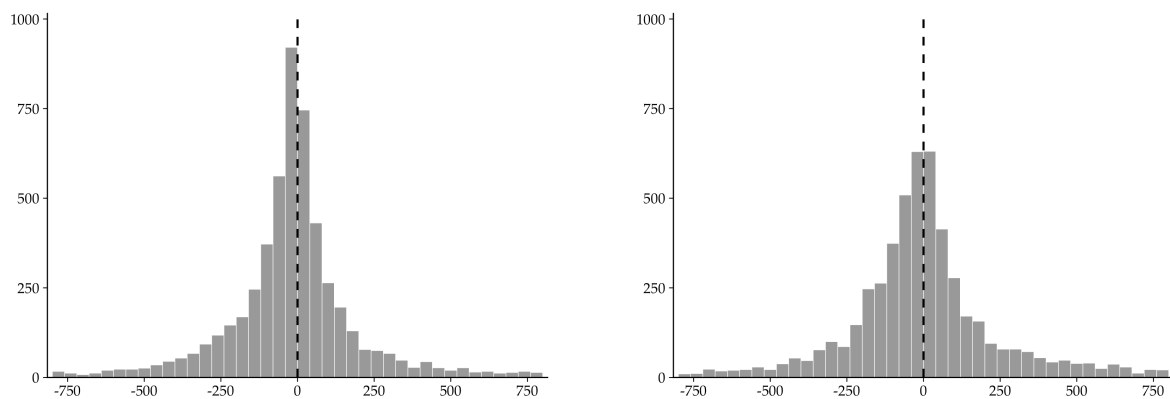


Note. Each dot is a municipality. The horizontal axis shows the federally authorized quota and the vertical axis shows the number of families actually enrolled in *Bolsa Família*, both in thousands. The dashed line marks the 45-degree locus of full take-up (realized = authorized). In December 2008, most municipalities lie close to or slightly below the 45-degree line, indicating that authorizations were approximately binding. In December 2009, following the federal quota reform, a substantial mass of municipalities shifts to the right of the 45-degree line, reflecting the expansion of authorized quotas that had not yet been translated into realized enrollment. The gap between the 45-degree line and the scatter in 2009 is the slack that our instrument exploits. Both axes are capped at the 99th percentile of the cross-municipal distribution (approximately 20,000 families); the small number of large urban municipalities above this threshold are excluded from the frame but not from any analysis.

This priority index considers each municipality's ratio of current beneficiaries to its

quota and the number of eligible-but-unserved families, creating a ratchet that, over time, tends to bring enrollment in line with the quota. The key implication is that the quota is neither a guarantee nor a purely notional figure: it constrains enrollment when binding, yet effective coverage also depends on the administrative initiative of local governments, creating a discretionary margin where political will and organizational capacity determine how many eligible families actually receive benefits.

Figure 3: Distribution of Slack Between Authorized Quota and Realized Enrollment



Note. Each panel shows the distribution of slack, defined as the authorized quota minus realized enrollment, across municipalities in December 2009, immediately following the federal reform. The left panel uses the official authorized quota; the right panel uses the counterfactual quota. Positive values indicate that a municipality received an authorization it had not yet translated into realized enrollment. The dashed vertical line marks zero slack. Both distributions are trimmed at the 5th and 95th percentiles to limit the influence of extreme observations. The mass of positive values in both panels confirms that the reform expanded quotas beyond contemporaneous enrollment levels for a substantial share of municipalities.

In 2009, the federal government revised the methodology used to compute municipal quotas, producing a heterogeneous reallocation of authorized beneficiary slots across municipalities. The previous approach, in place since 2006, distributed slots in proportion to each municipality's estimated number of poor families from the 2000 Census, adjusted for intercensal population growth. The 2009 revised methodology accounts for income volatility among families near the eligibility line, a feature of transient poverty that earlier projections systematically undercount ([Gerard et al., 2025](#)).

Alongside the methodological update, all municipal quotas were scaled by a factor of 1.18 to account for structural income volatility near the poverty threshold. Because this multiplicative factor is constant, it contributes no cross-municipal variation to the instrument; all identifying variation in the quota shock derives from the updated municipal poverty estimates and the change in income thresholds. Together, these changes added nearly 1.6 million new authorized slots in 2009. The reform expanded the ceiling, not the registry: although already-registered eligible families were absorbed auto-

matically, the magnitude of the expansion left a large residual margin of unfilled slots whose realization depended on the municipal government's effort to identify, locate, and register eligible families not yet in the federal database.

2.2 Data Sources

Our empirical analysis combines four municipal-level data sources: administrative records on *Bolsa Família*, electoral returns, census and household-survey data, and administrative records on local social-assistance infrastructure.

We begin with monthly administrative records from the *Ministério do Desenvolvimento Social* on *Bolsa Família* quotas, realized coverage, and total transfers from 2004 to 2016. Our main analysis uses the 2008 and 2012 windows to construct the treatment and instrument variables; the earlier (2004) and later (2016) years are reserved for placebo and persistence robustness exercises reported in the appendix. These data allow us to measure both authorized and realized program expansion.

We then use electoral records from the *Tribunal Superior Eleitoral* (TSE) covering municipal elections from 2000 to 2016. These data provide candidate-level vote totals, turnout, and detailed candidate characteristics, including education, prior occupation, and party affiliation.

To reconstruct counterfactual municipal quotas under the pre-reform allocation rule, we combine the 2000 Population Census from the *Instituto Brasileiro de Geografia e Estatística* (IBGE) with *Pesquisa Nacional por Amostra de Domicílios* (PNAD)-based estimates. These sources provide the inputs needed to approximate the quota formula that would have prevailed absent the reform.

Finally, we use administrative records from the *Censo SUAS* covering the universe of CRAS units in 2008 and 2009. In addition to information on the identity and tenure of directors and staff, these data include measures of local infrastructure and service provision, such as whether units conduct interviews and active searches, as well as the size of the population they serve.

2.3 Descriptive Statistics

The sample consists of 2,105 Brazilian municipalities observed in the 2008 and 2012 elections, those where the 2008 incumbent sought reelection. Table 1 reports descriptive statistics for the full population of municipalities and for this regression sample.

Three features stand out. The 2008 winner’s vote share averages 56.3 percent in the full population and 52.9 percent in the regression sample; the tighter dispersion reflects the rerunning restriction, which selects against incumbents at the extremes of the 2008 outcome distribution. *Bolsa Família* enrollment and the quota shock are both heavily right-skewed (BF mean 1,936 vs. median 947 in 2009), so both enter the regressions standardized. Census 2000 covariates have nearly identical means across panels, indicating the rerunning restriction does not select on the baselines we control for.

Table 1: Descriptive Statistics

	N	Min	Mean	Median	Max	SD
Panel A: All Municipalities						
Incumbent Won (2012)	4,730	0.00	55.35	100.00	100.00	49.72
Winner Vote Share (2008) (pp)	11,008	20.44	56.26	53.95	100.00	13.98
Realized Additional BF Expansion (2009)	11,008	-9,845.50	146.41	25.25	34,402.17	889.31
Excess Additional Beneficiaries (2009)	11,008	-5,282.54	378.03	144.65	113,606.29	1,990.32
Population (2000)	11,008	795.00	30,061.56	10,313.00	10,277,518.00	181,976.90
Gini (2000) (%)	11,008	30.00	54.73	55.00	87.00	6.82
Poor Population (2000) (%)	11,008	5.17	63.96	65.77	99.00	20.69
Income per capita (2000) (R\$)	11,008	62.65	338.02	308.11	1,759.76	192.21
Formal Workers Population (2000) (%)	11,008	1.92	36.07	34.44	86.38	18.13
Urban Population (2000) (%)	11,008	0.00	58.88	59.53	100.00	23.35
Panel B: Regression Sample						
Incumbent Won (2012)	4,210	0.00	56.34	100.00	100.00	49.60
Winner Vote Share (2008) (pp)	4,210	24.06	52.92	52.69	99.16	9.41
Realized Additional BF Expansion (2009)	4,210	-889.25	145.41	27.83	34,402.17	947.24
Excess Additional Beneficiaries (2009)	4,210	-5,282.54	349.80	145.78	50,052.59	1,446.66
Population (2000)	4,210	795.00	27,876.69	10,556.00	5,773,130.00	145,264.20
Gini (2000) (%)	4,210	30.00	54.84	55.00	80.00	6.73
Poor Population (2000) (%)	4,210	7.66	64.10	66.78	99.00	20.97
Income per capita (2000) (R\$)	4,210	69.20	337.21	300.12	1,759.76	195.03
Formal Workers Population (2000) (%)	4,210	2.89	35.78	33.99	86.38	18.28
Urban Population (2000) (%)	4,210	0.00	58.77	59.20	100.00	23.30

Notes: Panel A covers the universe of municipalities with a 2008 winner vote share and complete Census 2000 baseline controls; Panel B is the regression sample, municipalities where the 2008 incumbent sought reelection, subject to the data availability. Realized Additional BF Expansion is the muni-level inter-election change in the twelve-month moving average of beneficiary families between 2008 and 2012. Excess Additional Beneficiaries is the difference between the 2009 federal quota and the counterfactual quota, the quota that would have prevailed under the pre-reform allocation rule. The large standard deviation of this variable relative to its mean reflects substantial right skew driven by large urban municipalities; the regressions use a standardized version. The SD is smaller in Panel B than in Panel A because the regression sample drops municipalities without an incumbent rerunner. Full distributional statistics are reported in Appendix B.3.

3 Conceptual Framework

This section develops a simple framework for the political returns to welfare expansion in low-income settings like the one we study. The framework has two moving parts: a

multiplicative implementation chain in which federal authorization can only translate into realized expansion when the local administrative apparatus is in place, and a two-channel voter response in which beneficiaries reward the program materially and all voters update on the incumbent’s competence from observed implementation.

We proceed in three steps. The benchmark is a standard political-agency model in which local capacity is a single smoothly distributed index. The augmented model decomposes capacity into two institutionally grounded margins, physical infrastructure (the local CRAS apparatus) and bureaucratic alignment of the front-line manager, combined multiplicatively; the headline first stage is essentially identified where both margins are active. We close with a voter-response block that delivers the linear vote-share form and a partisan-attribution extension under which the inference channel sharpens when the mayor’s party also operates the federal program. The framework gives a parsimonious structure for interpreting the heterogeneity exercises, the partisan asymmetry, and the mechanism analysis that follows.

3.1 Agents

There are three classes of agents: a federal government, a continuum of municipalities indexed by m , each governed by an incumbent mayor, and a continuum of voters in each municipality indexed by i . Time runs over two periods, $t \in \{0, 1\}$, corresponding to the 2008 and 2012 municipal elections. The 2009 reform of *Bolsa Família*’s allocation formula occurs between the two periods.

Federal government. The federal government is non-strategic. It sets two policy parameters that are exogenous from the perspective of municipalities. First, it sets a per-family transfer value $\bar{b}$, uniform across municipalities and possibly differing between $t = 0$ and $t = 1$.⁶ Second, it sets a beneficiary quota Q_m^t for each municipality, computed from a national allocation rule based on census poverty estimates and intercensal population growth. We treat Q_m^t as exogenous and write the inter-election quota shock as

$$\Delta Q_m \equiv Q_m^1 - Q_m^{1,cf} \quad (1)$$

⁶Strictly, $\bar{b}$ summarizes a federally fixed schedule that pays a basic benefit to extremely poor families plus variable supplements per child under 16 and per adolescent aged 16–17. The schedule is uniform across municipalities and not subject to mayoral discretion, so $\bar{b}$ can be read either as a family-level transfer under a representative composition or as the population average under the local demographic distribution. Neither reading changes what follows, since the mayor cannot alter the schedule.

where $Q_m^{1,cf}$ is the counterfactual quota that would have prevailed under the pre-reform allocation rule.

Mayor. The incumbent mayor in municipality m inherits an administrative capacity K_m^0 fixed before period $t = 0$, parameterized by a legacy floor $\psi_0 \geq 0$, the passive utilization that operates through informal channels and prior administrative pathways even without a dedicated apparatus, and a ceiling $\bar{\psi} > 0$ that an idealized CRAS at full operational intensity would add on top of the floor.

Conditional on this inherited capacity, the mayor chooses discretionary effort $a_m \geq 0$ (active outreach by welfare staff, opening additional service points, mobilizing the registration apparatus, and pushing data updates through the federal pipeline) at a marginal cost that depends on his unobserved type $\theta_m \in \{0, 1\}$ representing competence, in the political-agency tradition surveyed by [Ashworth \(2012\)](#):

$$c(\theta_m) = c_0 - c_1 \cdot \theta_m, \quad c_0 > c_1 > 0. \quad (2)$$

Type is drawn by nature, observed by the mayor, and unobserved by voters; competent mayors thus face a lower marginal cost of effort and choose strictly higher a_m in equilibrium.

Voters. Voters care about two things: whether they receive the transfer, and whether the incumbent is competent.⁷ The material channel reflects the documented response of beneficiaries to direct programmatic transfers in cash-transfer settings ([Manacorda et al., 2011](#); [Zucco, 2013](#); [Avis et al., 2018](#)). The competence channel follows the retrospective-voting tradition: voters use observed government performance to update their beliefs about an unobserved politician type. Voter i 's utility differential from supporting the incumbent versus the challenger is

$$\Delta U_{im} = \alpha d_{im} + \delta \Pr(\theta_m = 1 \mid s_{im}) + \eta_{im}, \quad (3)$$

where $d_{im} \in \{0, 1\}$ indicates whether household i in municipality m is a beneficiary, s_{im} is a noisy signal of program implementation, $\alpha > 0$ is the material reward to receiving the transfer, $\delta > 0$ is the value voters place on competent incumbents, and η_{im} is an

⁷The model is agnostic about voter sophistication in separating federal from local contributions. The local component of delivery is directly observable: voters see CRAS operating, *busca ativa* happening, and registrations being processed locally. Whether voters additionally separate the federal component (benefit value, set by decree) is immaterial for the electoral effect: at most, a less sophisticated voter would also misattribute federal contributions, but the locally observable activity remains the proximate trigger.

idiosyncratic ideology shock.

Aggregate vote share. Aggregating individual choices across the continuum of voters under probabilistic voting in the tradition of [Lindbeck and Weibull \(1987\)](#), with the idiosyncratic shocks η_{im} smoothing across the ideology distribution, the muni-level vote share for the incumbent takes the linear form

$$V_m = V_0 + \lambda_1 B_m + \lambda_2 e_m, \quad (4)$$

where $\lambda_1 > 0$ aggregates the material channel, with each beneficiary reducing resistance to supporting the incumbent through α in equation (3), and $\lambda_2 \geq 0$ aggregates the inference channel, with visible implementation raising the posterior $\Pr(\theta_m = 1)$ for all voters, beneficiaries and non-beneficiaries alike.

3.2 Benchmark Model: Single Capacity Index

In the benchmark, capacity K_m^0 is summarized by a scalar $\psi_m \in [0, \infty)$, the inherited capacity that translates into effort-free quota utilization. Realized program coverage is

$$e_m = \psi_m + a_m, \quad B_m = e_m \cdot Q_m, \quad (5)$$

where e_m is the quota utilization rate and B_m is the number of realized beneficiaries.

The mayor's objective combines the electoral payoff, the convex cost of discretionary effort, and an institutional compliance pressure that penalizes deviations from full quota utilization, reflecting the federal monitoring of municipal performance against the authorized ceiling:

$$U_m = V_m - \frac{1}{2}c(\theta_m)a_m^2 - \frac{1}{2}\tau(e_m - 1)^2, \quad (6)$$

where V_m is the muni-level vote share from equation (4) and $\tau > 0$ parameterizes the enforcement pressure to remain near the quota benchmark.

Proposition 1 (Optimal Effort and Realized Expansion). *In the benchmark model, the unique optimal discretionary effort is*

$$a_m^*(\theta_m) = \frac{(\lambda_1 Q_m + \lambda_2) + \tau(1 - \psi_m)}{c(\theta_m) + \tau}. \quad (7)$$

Equilibrium realized coverage and beneficiaries are

$$e_m^*(\theta_m) = \psi_m + a_m^*(\theta_m), \quad B_m^*(\theta_m) = e_m^*(\theta_m) \cdot Q_m. \quad (8)$$

The first stage of the empirical design satisfies

$$\frac{\partial B_m^*}{\partial Q_m} = e_m^*(\theta_m) + Q_m \cdot \frac{\lambda_1}{c(\theta_m) + \tau} > 0 \quad (9)$$

for all municipalities with $\psi_m \geq 0$, $Q_m > 0$, and $a_m^* > 0$.

Proof. The argument is straightforward for the attentive reader: standard first-order conditions in a_m yield the expression above. Full details are presented in Appendix F. ■

The benchmark model predicts that the federal quota shock expands realized enrollment in every municipality, with the magnitude of expansion varying smoothly with capacity ψ_m , the size of the quota Q_m , and the mayor's competence θ_m . Heterogeneity in the first stage is therefore a matter of degree rather than of kind: the partial derivative $\partial B_m^*/\partial Q_m$ is strictly positive in every municipality with $\psi_m \geq 0$, $Q_m > 0$, and $a_m^* > 0$, so within-subgroup first stages cannot collapse to zero. The benchmark cannot accommodate a partition in which one subgroup's first stage approaches zero while another remains strongly identified, since $\partial B_m^*/\partial Q_m$ remains bounded below by the positive effort-response term $Q_m \cdot \lambda_1/[c(\theta_m) + \tau]$ even at the corner $\psi_m = a_m^* = 0$. Combined with $\lambda_1 > 0$ in equation (4), the model also predicts a positive headline 2SLS coefficient on incumbent vote share.

3.3 Augmented Model: Multiplicative Capacity

The benchmark imposes a strong restriction: local capacity is a single smooth scalar. The personnel-economics-of-the-state literature (Finan et al., 2017; Akhtari et al., 2022; Colonnelli et al., 2020) pushes back against this: local administrative output depends jointly on physical infrastructure and on the political incentives of the front-line staff who operate it. We therefore enrich the benchmark by allowing capacity to factorize along two institutionally grounded margins.

Infrastructure component. We write the first capacity component as $\kappa_m^I \in \{0, 1\}$ indicating whether municipality m had a *Centro de Referência de Assistência Social* (CRAS) in 2008. The CRAS is a ready-made bureaucratic apparatus for enrolling households

into social programs, with permanent staff, standardized protocols, and a direct data pipeline into the *Cadastro Único*. Finan et al. (2017) place this kind of pre-existing local infrastructure at the heart of the state-capacity literature: front-line administrative output is shaped by the agencies and personnel a municipality has on hand to execute federal mandates. Municipalities without a CRAS can still enroll new beneficiaries through alternative channels (public hospitals, mobile *cadastro* teams, ad hoc active outreach), but the productivity of those channels is lower than what an existing CRAS delivers. The same federal authorization therefore translates into less realized expansion where the apparatus is missing, and more where it is in place.

Alignment component. We write the second capacity component as $\kappa_m^A \in \{0, 1\}$ indicating whether the CRAS manager (the *coordenador* of the unit) was replaced after the 2008 municipal election, when the newly elected mayor took office. Two independent literatures motivate the relevance of front-line manager replacement as a productivity margin. Akhtari et al. (2022) document that mayoral transitions in Brazil systematically trigger turnover of street-level bureaucrats, with replacements concentrated in positions where the mayor has hiring discretion, including managers of social-assistance and education facilities. Colonnelli et al. (2020) document, in the same Brazilian setting, that public-sector positions function as electoral stepping stones, with their attractiveness as career-launching pads tied to the operational footprint of the office. Both readings imply that a post-2008 manager change can shift the productivity of the implementation chain: under the first, through replacement with bureaucrats more responsive to the incoming mayoral coalition, and under the second, through replacement with politically ambitious agents who exert more effort because program expansion raises the visibility of the post and its attractiveness as a launching pad for a local electoral career. The model treats κ_m^A as a generic productivity margin consistent with both readings.

Production with multiplicative capacity. The implementation chain that translates federal authorization into realized expansion has two indispensable capacity components: an existing registration apparatus and an active front-line manager. Kremer (1993) and Jones (2011) characterize technologies of this form: the productivity of the chain is the product of component qualities, so the failure of any single component governs the productivity of the whole.

We adopt this structure with a productivity floor $\gamma \in [0, 1)$ to allow attenuation

rather than literal collapse when one capacity component is weak. Let

$$g(\kappa_m^I, \kappa_m^A) = \gamma + (1 - \gamma) \kappa_m^I \kappa_m^A, \quad \gamma \in [0, 1], \quad (10)$$

so that $g = 1$ when both capacity components are active and $g = \gamma$ otherwise. Realized coverage takes the form

$$e_m = \psi_0 + g(\kappa_m^I, \kappa_m^A) \cdot (\bar{\psi} + a_m), \quad B_m = e_m \cdot Q_m, \quad (11)$$

where ψ_0 and $\bar{\psi}$ are the floor and ceiling defined above. The strict Leontief case is the limit $\gamma = 0$; with $\gamma > 0$, subgroups that lack one capacity component still exercise some quotas, just at the attenuated rate g .

The mayor's objective remains as in equation (6), with V_m now driven by the augmented e_m and B_m from equation (11). Solving yields the following characterization.

Proposition 2 (Optimal Effort and Attenuated First Stage). *In the augmented model with production (11) and multiplier (10), optimal discretionary effort $a_m^*(\theta_m)$ is strictly increasing in the multiplier $g(\kappa_m^I, \kappa_m^A)$. The first stage satisfies*

$$\frac{\partial B_m^*}{\partial Q_m} = e_m^*(\theta_m) + Q_m \cdot \frac{\lambda_1 g^2}{c(\theta_m) + \tau g^2}, \quad (12)$$

which is strictly increasing in g . As $\kappa_m^I \kappa_m^A \rightarrow 0$ the first stage attenuates by a factor of order γ^2 relative to the active subgroup. The attenuation operates on the marginal-productivity term and propagates through equilibrium effort.

Proof. The argument is straightforward for the attentive reader: standard first-order conditions in a_m combined with the chain rule applied to production (11) yield the expression above. Full details are presented in Appendix F. ■

The test against the benchmark is sharp on signs and orders of magnitude. The static-response component of equation (12) attenuates linearly in γ through $g\bar{\psi}$, while the effort-response component attenuates more steeply, by γ^2 , because the multiplier enters the effort margin twice: a weaker chain dampens how much each unit of effort raises coverage, and it also blunts the mayor's incentive to exert effort in the first place. Under the assumption that ψ_0 is small relative to $\bar{\psi}$, the ratio of inactive- to active-subgroup first-stage slopes is bounded between γ and γ^2 . A within-subgroup first-stage ratio close to one rejects the augmented model; any substantially below one is consistent with it and provides a range estimate of γ .

The multiplier $g(\kappa_m^I, \kappa_m^A) = \gamma + (1 - \gamma) \kappa_m^I \kappa_m^A$ treats the two capacity margins symmetrically: a single γ governs the attenuation when either component is inactive. The

framework therefore makes no prediction that infrastructure and alignment must attenuate at the same rate; if the data return distinct γ_{κ^I} and γ_{κ^A} , this should be read as an empirical refinement consistent with the framework rather than as an additional model prediction.

The augmented model preserves the benchmark’s headline-level predictions, both a positive first stage on the full sample and a positive 2SLS coefficient on vote share, but predicts a sharply different shape of heterogeneity. Where the benchmark predicts smooth attenuation across subgroups, the augmented model predicts that the first stage essentially collapses where any structural margin is missing. In municipalities without a pre-existing *CRAS*, the implementation chain has no apparatus to translate authorized slots into realized expansion. Inside *CRAS* municipalities, the same logic applies to manager turnover: where the front-line operator was not replaced after the 2008 municipal election, the chain operates at the floor productivity γ rather than at full intensity. The active cell, where both margins are simultaneously present, is where the headline 2SLS coefficient is essentially identified.

3.4 Voter Response

We close the model by characterizing how realized expansion translates into vote share. Voters do not observe the mayor’s type θ_m directly; instead, they form beliefs about competence from observable program implementation outcomes, in the spirit of the retrospective-voting tradition of [Ferejohn \(1986\)](#) and the more recent literature on voters learning from government performance in developing democracies ([Ferraz and Finan, 2008](#); [Avis et al., 2018](#)). Each voter i in municipality m observes a noisy signal

$$s_{im} = \nu_0 + \nu_1 d_{im} + \nu_2 e_m + \varepsilon_{im}, \quad \nu_1, \nu_2 > 0, \quad (13)$$

where ε_{im} is mean-zero noise independent across voters and conditional on e_m . The first informative component, $\nu_1 d_{im}$, captures the voter’s direct experience as a beneficiary. The second, $\nu_2 e_m$, captures aggregate visibility of program delivery (signage, neighborhood-level enrollment campaigns, local media coverage).

Bayesian updating closes the model in the standard career-concerns and political-agency tradition of [Holmström \(1999\)](#) and [Maskin and Tirole \(2004\)](#), surveyed by [Ashworth \(2012\)](#). Voters apply Bayes’ rule using the priors $p_0 = \Pr(\theta_m = 1)$ and the conditional signal densities f_0 and f_1 :

$$\Pr(\theta_m = 1 \mid s_{im}) = \frac{f_1(s_{im})p_0}{f_1(s_{im})p_0 + f_0(s_{im})(1 - p_0)}. \quad (14)$$

By Proposition 1, competent mayors choose strictly higher a_m^* than incompetent mayors and therefore generate higher e_m^* . This shifts the conditional mean of s_{im} upward under $\theta_m = 1$. For a wide class of noise distributions (specifically, those with log-concave densities), this location shift implies the monotone likelihood ratio property:

$$\frac{f_1(s)}{f_0(s)} \text{ is strictly increasing in } s. \quad (15)$$

Under (15), the posterior in (14) is strictly increasing in s_{im} . Voter inference is rational: a higher signal does indicate higher probability of competence, because in equilibrium higher signals are more likely to have been generated by competent mayors. Aggregating across voters in the probabilistic-voting tradition of Lindbeck and Weibull (1987), with η_{im} smoothing across the ideology distribution, the linear vote-share form posited in equation (4) is recovered as the equilibrium aggregate. The aggregation parameter $\lambda_1 > 0$ collects the material channel (αd_{im} across beneficiaries), and $\lambda_2 \geq 0$ collects the inference channel ($\delta \Pr(\theta_m = 1 \mid s_{im})$ across all voters, beneficiaries and non-beneficiaries alike).

Remark 1 (The two channels and the sign of the headline). *Equation (4) predicts $\partial V_m / \partial Q_m > 0$ whenever $\partial B_m^* / \partial Q_m > 0$ (Proposition 1 or 2) and either $\lambda_1 > 0$ or $\lambda_2 > 0$. The sign of the headline 2SLS coefficient therefore tests jointly whether the first stage operates and whether voters reward implementation.*

Partisan attribution and the inference channel. The inference parameter λ_2 aggregates the rate at which voters convert observed implementation into a posterior on the mayor's competence. A natural extension lets λ_2 vary across municipalities with the alignment between the local executive's partisan brand and the federal program's institutional sponsor. When the mayor's party is also the federal incumbent that designed and operates the program at the national level, the credit chain runs cleanly from local registration to national policy under a single partisan label; voters' Bayesian update on competence is sharper because the inferred type controls both ends of the implementation chain. Formally, decompose the inference parameter as $\lambda_{2,m} = \lambda_2^{\text{base}} + \lambda_2^{\text{attrib}}$. $\mathbf{1}\{\text{partisan alignment}_m\}$, with $\lambda_2^{\text{attrib}} \geq 0$. The augmented vote-share function becomes

$$V_m = V_0 + \lambda_1 B_m + \lambda_{2,m} e_m, \quad (16)$$

and the per-beneficiary electoral return is larger under aligned mayors by $\lambda_2^{\text{attrib}}$. The model thus predicts that, holding realized expansion fixed, a municipality governed by a mayor whose party owns the federal program returns a higher per-beneficiary vote share than one governed by a non-aligned mayor. This prediction is testable on the

partition that separates Workers’ Party mayors from the rest, under the identifying restriction that the material channel λ_1 is partisan-invariant; the restriction is institutional, since the federally set benefit value and eligibility rules make the marginal contribution of a beneficiary to αd_{im} independent of local partisan alignment. The partisan multiplier is therefore a feature of the model, not an unmodeled regularity.

4 Empirical Strategy

Our empirical strategy exploits a 2009 federal reform that revised the formula used to compute municipal *Bolsa Família* quotas. The reform generated plausibly exogenous variation in municipalities’ authorized beneficiary ceilings, but turning that authorization into actual enrollment depended on local implementation. The wedge between what each municipality was allowed to do and what it actually did is what lets us identify the political return to expansion. The analysis is a difference-in-differences design combined with two-stage least squares, and it uses the federally induced quota expansion as an instrument for realized changes in beneficiary enrollment.

4.1 The 2009 Quota Reform as Source of Identification

The methodology for computing municipal quotas was implemented by the *Ministério do Desenvolvimento Social* (MDS) based on census and survey data, following a standardized algorithm with no discretion at either the local or federal executive level. [Brollo et al. \(2020\)](#) provide direct evidence supporting this claim. They find evidence of manipulation in the enforcement of program conditionalities but no indication of political influence over the allocation of quotas themselves.⁸

To isolate the variation attributable solely to the change in allocation rules, we construct counterfactual 2009 quotas by applying the pre-reform methodology, which had been used to set the 2004 and 2006 quotas, to the data that would have been available immediately before the reform,

$$\text{CountQuota}_m^{2009} = \frac{\text{Poor}_m^{2000} \cdot n_m^{[2000,2006]}}{\sum_{k \in s} \text{Poor}_k^{2000} \cdot n_k^{[2000,2006]}} \cdot 1.18 \cdot \text{Poor}_s^{2006} \quad (17)$$

⁸[Gerard et al. \(2025\)](#) note that they cannot exactly reproduce the official MDS quota numbers using publicly available information and the pre-reform allocation formula. They do not interpret this discrepancy as evidence of political manipulation, and a more plausible explanation is that the remaining differences reflect unpublished administrative adjustments or small differences in auxiliary estimates used to operationalize the formula.

where $Poor_m^{2000}$ is the number of poor families in municipality m of state s from the 2000 Census, $n_m^{[2000,2006]}$ is an IBGE estimate of municipal population growth between 2000 and 2006, $Poor_s^{2006}$ is the state-level estimate of poor families from the 2006 *Pesquisa Nacional por Amostra de Domicílios* (PNAD), and the factor 1.18 reflects a uniform expansion applied to all municipal quotas as part of the reform.⁹ The actual 2009 quotas were computed by MDS using a revised methodology that incorporated updated poverty estimates and new income-volatility adjustments, and the exact formula is not publicly available. The preferred instrument is therefore

$$\Delta Quota_m^{2009} = Quota_m^{2009} - CountQuota_m^{2009} \quad (18)$$

This measure captures the cross-municipal variation in authorized expansion arising from the change in the federal allocation methodology, net of the uniform national expansion and local poverty differences. In the preferred specification, the counterfactual quota counts as poor any household with monthly per-capita income below R\$90 in the 2000 Census and below R\$120 in the 2006 PNAD. These were the official *Bolsa Família* eligibility thresholds in force during the pre-reform allocation rounds of 2004 and 2006.¹⁰ The 2009 reform announcement (MDS *Informe nº 169*) introduced an expanded R\$137 cutoff alongside the formula change. Using R\$137 inside the counterfactual would mix two distinct sources of variation in $\Delta Quota_m$, the change in formula (the policy-relevant variation we want to instrument with) and the mechanical change in the count of poor families that follows from a higher cutoff applied to the same PNAD micro-data, which is a definitional change unrelated to the formula's distributional logic. The R\$90/120 counterfactual instead holds the data definitions fixed at their pre-reform values, so $\Delta Quota_m$ isolates the formula component.¹¹

⁹The last officially reported **quota update** before the 2009 reform used information from the 2000 Census and the 2004 PNAD. To construct the counterfactual, we update the pre-reform formula using 2006 PNAD data, which is the most recent information that would have been available to policymakers at the time, while holding the allocation rule fixed.

¹⁰Validation against the official 2004 and 2006 quotas confirms that the R\$90 cutoff applied to the 2000 Census and the R\$120 cutoff applied to the 2006 PNAD jointly produce the closest replication of the actual pre-reform quota allocations. See Appendix B for details.

¹¹The R\$90/137 alternative is not just conceptually less clean but also empirically less stable. Under that construction, the leave-one-state-out exercise reveals identifying mass concentrated in Rio de Janeiro, and dropping that state collapses the Kleibergen-Paap F -statistic well below conventional thresholds. The preferred R\$90/120 instrument is stable in this dimension and across every other sample perturbation. Appendix C.3 documents the comparison.

4.2 Research Design

The design compares changes in incumbent electoral performance between 2008 and 2012 across municipalities that faced different intensities of authorized quota expansion. Identification follows the continuous-treatment IV framework of [Angrist and Imbens \(1995\)](#), which extends the binary-instrument LATE setup of [Imbens and Angrist \(1994\)](#) to instruments and treatments with multiple ordered values. Three assumptions carry the argument, and we discuss each in detail below. The first is conditional independence of the quota shock with respect to potential electoral outcomes. The second is the exclusion restriction, which says that the shock affects vote share only through realized beneficiary expansion. The third is monotonicity of the first stage. When all three hold, the residual variation in ΔQuota_m after the fixed effects and baseline-by-post controls isolates the causal effect of discretionary program expansion on subsequent electoral outcomes. We have only two election periods, which rules out formal pre-trend testing, so we probe conditional independence through a proportional-selection exercise that complements the placebo and balance tests that follow.

The primary challenge is the endogeneity of realized enrollment. Mayors influence how many families are actually registered, so coverage growth reflects local political incentives. Municipalities where incumbents face stronger electoral pressures may expand coverage more aggressively precisely because the expected political return is higher ([Pereira et al., 2020](#)). We address this by using the exogenous federal authorization shock as an instrument for realized beneficiary expansion. The data are organized as a balanced municipality-by-election-year panel covering the 2008 and 2012 elections, and we estimate the system

$$\text{Beneficiaries}_{mt} = \alpha_m + \lambda_t + \pi_1 (\Delta\text{Quota}_m \times \text{Post}_t) + X'_{mt}\pi_2 + \eta_{mt} \quad (19)$$

$$\text{VoteShare}_{mt} = \alpha_m + \lambda_t + \beta_1 \widehat{\text{Beneficiaries}}_{mt} + X'_{mt}\beta_2 + \varepsilon_{mt} \quad (20)$$

where α_m and λ_t are municipality and year fixed effects, Post_t is an indicator for the 2012 election, and X_{mt} bundles the incumbent-party fixed effects and the Census 2000 baseline characteristics interacted with Post_t . The variable $\text{Beneficiaries}_{mt}$ is the twelve-month moving average of enrolled families in municipality m measured at election year t . For $t = 2008$ the average runs over the twelve months of 2008, and for $t = 2012$ it runs over the twelve months of 2009, the year in which the quota shock materialized and induced the policy-relevant expansion.¹² The instrument ΔQuota_m is the time-

¹²Alternative definitions extending the beneficiary measurement window through 2012 yield similar

invariant 2009 quota shock, and it enters as $\Delta\text{Quota}_m \times \text{Post}_t$ because the shock itself is a single number per municipality and would otherwise be absorbed by the municipality fixed effects.

Equations (19) and (20) describe the data-generating process at the panel level, but the estimator we implement reads more transparently in first differences. With two periods, the municipality fixed effects α_m are time-invariant within municipality and drop out of the within-municipality demeaning, leaving the inter-election change as the operative variation. The system reduces to a cross-sectional IV at the municipality level,

$$\Delta\text{VoteShare}_m = \Delta\lambda + \beta_1 \Delta\widehat{\text{Beneficiaries}}_m + \Delta X'_m \beta_2 + \Delta\varepsilon_m \quad (21)$$

where $\Delta\text{Beneficiaries}_m$ is the change in the twelve-month moving average of enrolled families between the 2008 and 2012 elections and $\Delta\lambda$ is the difference in year fixed effects, a common time trend that absorbs whatever shifted electoral outcomes nationally between the two elections. The coefficient β_1 identifies the marginal effect of the muni-level change in beneficiaries between elections on the corresponding change in vote share, with the quota shock ΔQuota_m instrumenting the endogenous portion of $\Delta\text{Beneficiaries}_m$.

The two displays serve complementary purposes. Equations (19) and (20) state the structural model on the original panel, while equation (21) states the regression that delivers β_1 in practice.¹³ The two specifications coincide when $\text{Cov}(\text{Beneficiaries}_{m,2008}, \Delta\text{Quota}_m) = 0$, but in our setting that covariance is positive. The 2009 reform reweighted the formula toward the same dimensions of municipal poverty that already drove pre-reform coverage, so the post-interacted specification understates β_1 . The first-difference specification is the natural target when, as here, the substantive object is the political return to expansion rather than the political return to a higher 2012 level conditional on an unobserved 2008 baseline.

Both $\text{Beneficiaries}_{mt}$ and ΔQuota_m are standardized to mean zero and unit variance within the estimation sample, so coefficients in the tables read as the response to a one-standard-deviation move in the relevant variable. The two natural-unit scales that anchor the substantive interpretation are the muni-level “additional families” quantities reported in Panel B of Table 1. The cross-municipal standard deviation of the inter-election change in realized beneficiaries (the *Realized Additional BF Expansion* row) is 947 families, and the cross-municipal standard deviation of the 2009 quota shock (the

results in Appendix C.3.

¹³In the ANCOVA-style (McKenzie, 2012) alternative, the endogenous variable is interacted with Post_t and the first-difference reduction collapses to the 2012 level of beneficiaries rather than the inter-election change, and we discuss it in Appendix C.9.

Excess Additional Beneficiaries row) is 1,447 authorized families.

The first-difference IV in equation (21) therefore relates these two on a like-for-like scale, additional realized beneficiaries between elections instrumented by additional authorized beneficiaries from the 2009 reform, both measured at the municipality level. A 2SLS coefficient of one in standardized units corresponds to approximately 0.19 percentage points of vote share per typical inter-election expansion of 947 families, or about 0.20 percentage points per 1,000 additional beneficiaries enrolled between the two elections. The tables report estimates in standardized units, and the body text translates them into per-thousand-families terms.¹⁴

Treatment varies continuously in intensity rather than in binary status, so the design identifies an average causal response (ACR) parameter in the sense of Angrist and Imbens (1995). The ACR is a weighted average of the marginal effect of expansion on the change in vote share, where the non-negative weights $\omega(d)$ are proportional to $\text{Cov}(\mathbf{1}\{D_m \geq d\}, Z_m)$ and integrate to one. Under Angrist and Imbens’s assumptions of independence, exclusion, and monotonicity, β_1 is the population analog of $\int \omega(d) \mathbb{E}[\partial Y(d)/\partial d] dd$. Realized expansion is right-skewed, so these weights concentrate among municipalities where the 2009 reform translated into substantive enrollment changes, and the population weights further re-weight the estimand toward the experience of the typical Brazilian citizen rather than the typical municipality. Because no municipality is fully untreated in either period, the design does not identify level treatment effects (ATT or ATE), and β_1 is the marginal response, not the level response.¹⁵

Outcome. The main outcome is the incumbent’s vote share,

$$\text{VoteShare}_{mt} = \frac{\text{IncumbentVotes}_{mt}}{\text{VotesOnCandidates}_{mt}}$$

where $\text{VotesOnCandidates}_{mt}$ is the total number of valid votes cast for any candidate in municipality m in election year t , excluding blank and null ballots. This normalization ensures that the outcome reflects the incumbent’s competitive position among voters

¹⁴Standardization is an affine transformation that does not affect t -statistics, p -values, the Kleibergen-Paap first-stage F , or Anderson-Rubin confidence sets. $\text{Beneficiaries}_{mt}$ is standardized using panel-level dispersion (4,949 families), and the conversion factor to the inter-election scale is $947/4,949 \approx 0.19$. Appendix C.3 reports raw-unit results.

¹⁵The recent panel-DiD literature uses the same “ACR” label for related parameters identified under strong parallel-trends assumptions at every dose level (notably Callaway et al., 2024). We do not invoke this approach. Angrist and Imbens’s ACR is identified through instrument exogeneity and monotonicity rather than through dose-level parallel trends.

who expressed a preference for a candidate.¹⁶

Sample. The estimation sample is restricted to municipalities in which the incumbent mayor elected in 2008 sought reelection in 2012. After excluding municipalities where the incumbent did not seek reelection, those with missing treatment, outcome, or baseline covariate information, and those without contested elections in both periods, the final sample contains 2,105 of the 5,568 Brazilian municipalities (approximately 38 percent), observed in two election years and yielding 4,210 municipality-year observations. Because the outcome is defined at the incumbent level, this restriction is unavoidable. Section 4.3.1 shows that the instrument does not predict rerunning, and Table 2 confirms that the estimation sample is broadly representative of the full population of municipalities.

Specification. All specifications include municipality and year fixed effects. Because municipality fixed effects absorb time-invariant municipal characteristics, the baseline controls enter only as interactions with the post indicator. In our preferred specification we additionally include incumbent-party fixed effects based on the pre-reform winner's party, which absorb party-specific electoral dynamics, and a bundle of muni-level baseline characteristics from the 2000 Census interacted with post (log population, population growth, share of poor, household income per capita, and urban share), which absorb any differential 2012 effect operating through pre-reform municipal conditions. We exclude the incumbent's 2008 vote share from the preferred specification to avoid introducing a lagged dependent variable. Combining unit fixed effects with a lagged dependent variable in a short panel is known to generate bias (Nickell, 1981; Imai and Kim, 2019), and the lagged outcome appears as a sensitivity check in Section 5.4.

All regressions are weighted by municipal population in 2000. This is a scope condition rather than a precision choice. The estimand targets the causal response for the typical Brazilian citizen, which, given Brazil's urban concentration, weights toward larger municipalities where the absolute scale of expansion was economically meaningful.¹⁷

Standard errors are clustered at the state level, following Abadie et al. (2023), because the counterfactual quota formula incorporates a state-level poverty estimate and

¹⁶Appendix C.3 reports qualitatively unchanged results using alternative outcome definitions, including party vote share, and alternative denominators such as total votes cast and the eligible electorate.

¹⁷The right-skewed distribution of realized *Bolsa Família* expansion would otherwise give dominant weight to municipalities where expansion was modest and administrative capacity limited. Population weights align the estimand with the municipalities where the mechanism is most likely to operate. Unweighted and alternative weights' estimates are reported in Appendix C.3.

induces within-state correlation in the instrument.¹⁸

4.3 Threats to Identification

Identification of β_1 as the average causal response requires conditional independence, the exclusion restriction, and monotonicity. We discuss each in turn and document, where possible, that the data are consistent with the assumption. A separate concern about the timing of the 2012 quota revision is addressed at the end of the subsection.

4.3.1 Conditional Independence

Assumption 1 (Conditional Independence). *Conditional on municipality fixed effects, year fixed effects, incumbent-party fixed effects, and the baseline Census 2000 characteristics interacted with the post indicator, the quota shock is orthogonal to potential electoral outcomes:*

$$\begin{aligned} \mathbb{E}[Y_{m,t}(0) \mid \Delta\text{Quota}_m \times \text{Post}_t, \alpha_m, \lambda_t, \pi_m^{2008}, X_m^{2000} \times \text{Post}_t] \\ = \mathbb{E}[Y_{m,t}(0) \mid \alpha_m, \lambda_t, \pi_m^{2008}, X_m^{2000} \times \text{Post}_t]. \end{aligned}$$

The quota shock must be orthogonal to potential electoral outcomes conditional on municipality fixed effects, year fixed effects, incumbent-party fixed effects, and the baseline characteristics interacted with the post indicator. Conditional rather than unconditional exogeneity is the relevant requirement here because the counterfactual quota formula incorporates a state-level poverty estimate, Poor_s^{2006} , which mechanically correlates the instrument with pre-existing political geography. Municipalities in poorer states receive systematically different quota shocks, and those same states have structurally distinct electoral dynamics (Zucco, 2013). Once the controls absorb time-invariant municipal characteristics, party-specific electoral baselines, and any 2012 trend operating through pre-reform municipal conditions, the residual variation in ΔQuota_m reflects the change in the federal allocation formula net of these dimensions and is plausibly orthogonal to potential electoral outcome trends.

The proportional-selection exercise follows Oster (2019) and computes δ^* , the ratio of selection on unobservables to selection on observables that would be required to

¹⁸Brazil is divided into 26 states plus the Federal District. The Federal District does not hold municipal elections and falls outside our sample, so the regressions cluster on 26 state-level groups. Even at this count, asymptotic cluster-robust inference can be unreliable when the first stage is strong. We therefore report Anderson and Rubin (1949) confidence sets throughout, which are valid under arbitrary instrument weakness and robust to the cluster count. Appendix C.4 also reports wild cluster bootstrap p -values (Cameron et al., 2008; MacKinnon et al., 2023).

explain the reduced-form effect away under the framework’s proportional-selection assumption. Adding the Census 2000 baseline characteristics interacted with post moves the coefficient on the quota shock from 0.64 to 0.61. The decline is small and consistent with limited selection on observables. Under the conservative bound $\bar{R}^2 = 1.3 \cdot R_{\text{long}}^2$ we obtain $\delta^* = 24.12$, and under [Oster’s](#) recommended benchmark $\bar{R}^2 = 2.2 \cdot R_{\text{long}}^2$ we obtain $\delta^* = 6.03$. For proportional selection on unobservables to overturn the reduced-form effect, unobservables would have to be at least six times as important as the rich set of observables we condition on.

Selection into rerunning. The estimation sample is restricted to municipalities where the 2008 incumbent sought reelection in 2012, so the instrument could bias inference if it systematically affected the decision to run again. [Table 2](#) regresses an indicator for whether the 2008 incumbent appears on the 2012 ballot on the quota shock, estimated on the full universe of municipalities. Under the preferred counterfactual-based instrument, the estimated coefficients are small and statistically indistinguishable from zero. The identifying variation appears orthogonal to selection into the estimation sample.

Table 2: Instrument and the Incumbent’s Decision to Run Again in 2012

	Incumbent Run in 2012		
	(1)	(2)	(3)
$\Delta\text{Quota}_m(\hat{\gamma}_1)$	-0.005 (0.003)	-0.004 (0.005)	-0.003 (0.004)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	5,504	5,504	5,504

Notes: The dependent variable is an indicator equal to one if the 2008 incumbent appears on the ballot in the 2012 municipal election. The regression is estimated on the universe of municipalities, using the preferred instrument. Column (1) is bivariate; column (2) adds party fixed effects based on the 2008 winner; column (3) further adds baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Column (3) is the preferred specification, matching the control set of the main DiD-IV. Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000. 64 observations were excluded due to missing values in one or more variables.

As a complementary check, [Section 5.4](#) re-estimates the preferred specification weighted by the inverse probability of selection into the rerunning sample. The 2SLS point estimate moves from 2.94 to 2.80 and remains statistically significant, which indicates that selection into the rerunning sample is not materially distorting the

estimates.

Pre-reform electoral balance. A second exercise tests whether the quota shock is correlated with pre-existing political conditions. Table 3 reports placebo regressions of 2008 electoral outcomes on the preferred instrument within the estimation sample. Columns 1 to 3 use the 2008-winner’s vote share as the outcome, and columns 4 to 6 use an indicator for the sitting incumbent winning in 2008. Because 2008 outcomes were determined before the reform, any systematic non-zero association would signal baseline imbalance rather than the effect of quota-induced expansion.

Table 3: Placebo: 2008 Electoral Outcomes on Post-Reform Quota Shock

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{Quota}_m(\hat{\delta}_1)$	-0.026 (0.063)	-0.036 (0.066)	-0.110 (0.069)	-0.005 (0.004)	-0.015 (0.014)	-0.002 (0.013)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	2,105	2,105	2,105	2,105	2,105	2,105

Notes: The dependent variable is measured in the 2008 municipal election. Vote Share is the 2008-winner’s share of valid votes. Winning is an indicator equal to 100 if the sitting incumbent (2004-winner) won in 2008 and 0 otherwise. All columns use the preferred instrument and restrict to the municipalities in the DiD-IV estimation sample. Columns (1) and (4) are bivariate; columns (2) and (5) add party fixed effects based on the 2008 winner; columns (3) and (6) further add baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000.

The evidence is consistent with conditional orthogonality of the instrument. Point estimates remain near zero and statistically insignificant as progressively richer control sets are added, and none of the six specifications rejects the null of zero association at conventional levels. The instrument does not display the kind of pre-reform imbalance that would threaten identification.

No anticipation. The 2009 quota revision was formally announced in MDS *Informe n° 169*, dated April 2009, which communicated that approximately 300,000 additional families would be incorporated in staggered waves over the course of the year. The previous quota adjustment had occurred in 2006. Although municipalities may have anticipated that some revision would eventually take place, they could not have foreseen

the magnitude, the distributional formula, or the timing of the 2009 reallocation before the official announcement, which came six months after the October 2008 municipal elections. The placebo balance reported just above is consistent with this institutional timeline, since the instrument is statistically unrelated to either the 2008 winner’s vote share or the sitting incumbent’s 2008 win probability in any specification.

4.3.2 Exclusion Restriction

Assumption 2 (Exclusion Restriction). *The quota shock affects incumbent vote share only through realized beneficiary enrollment:*

$$\Delta \text{Quota}_m \times \text{Post}_t \perp\!\!\!\perp Y_{m,t}(\text{Beneficiaries}_{m,t}) \\ | \text{Beneficiaries}_{m,t}, \alpha_m, \lambda_t, \pi_m^{2008}, X_m^{2000} \times \text{Post}_t.$$

Equivalently, conditional on the controls and on realized expansion, the quota shock has no direct effect on the incumbent’s vote share.

The exclusion restriction states that the quota shock affects vote share only through realized beneficiary enrollment. This is fundamentally a substantive rather than a statistical claim, and the case for it rests on the institutional channel through which the shock operates. The 2009 reallocation was a federally administered formula change, computed centrally by the *Ministério do Desenvolvimento Social* from Census 2000 and PNAD 2006 data, and it modified one and only one variable observable to municipalities and to voters, namely the authorized number of beneficiary slots. The benefit value paid per family was set by separate federal decree, and conditionality enforcement was implemented by federal partners through schools and health facilities.

The institutional argument is the primary defense of the exclusion restriction, but the most plausible alternative channels have empirical counterparts that we can test directly. Two concurrent federal-to-municipal flows are worth addressing: the per-family *Bolsa Família* benefit value and the constitutionally fixed revenue-sharing transfer (*Fundo de Participação dos Municípios*, FPM). Both are institutionally decoupled from the quota formula. The mean per-family benefit value is set by national rules but varies in the cross section through family composition, which is in principle correlated with the quota shock through the same Census 2000 poverty estimates that feed the formula. FPM coefficients are set by Complementary Law 91/1989 from fixed population brackets and apportionment factors; they are not a function of poverty estimates, of the 2007 *Contagem* demographic update, or of the income-volatility correction that drove the 2009 *Bolsa Família* reweighting.

Table 4 reports the empirical exclusion tests. Panel A regresses the inter-election change in FPM transfers (interacted with post) on the standardized quota shock (interacted with post), under three control specifications matching the headline. Panel B reports the analogous test using the per-family *Bolsa Família* benefit value. In both panels, the preferred specification (column 3, with Census 2000 baseline characteristics interacted with post) returns a coefficient statistically indistinguishable from zero. The empirical evidence is consistent with the institutional argument: the quota shock does not differentially shift per-family benefit values or concurrent FPM transfers across municipalities, so neither channel absorbs non-enrollment variation that would contaminate the IV.

Table 4: Instrument Exogeneity: FPM Transfers and Benefit Value

	(1)	(2)	(3)
<i>Panel A: $\Delta FPM_m \times Post$ (std)</i>			
$\Delta Quota_m \times Post$ ($\hat{\gamma}$)	-0.265 (0.037)	-0.294 (0.032)	0.011 (0.060)
<i>Panel B: $\Delta Value_m \times Post$ (std)</i>			
$\Delta Quota_m \times Post$ ($\hat{\gamma}$)	-0.003 (0.003)	-0.006 (0.003)	0.003 (0.002)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	4,210	4,210	4,210

Notes: Each panel reports the coefficient on the standardized 2009 quota shock interacted with post in a regression where the dependent variable is the indicated muni-level constant interacted with post. Panel A uses the inter-election change in *Fundo de Participação dos Municípios* transfers ($\Delta FPM_m = FPM_m^{2009} - FPM_m^{2008}$, standardized). Panel B uses the twelve-month moving average of the per-family *Bolsa Família* benefit value (standardized). Both regressions are estimated on the DiD-IV sample of 4,210 municipality-year observations using the preferred R\$90/120 instrument. Column (1) includes only year and municipality fixed effects; column (2) adds incumbent-party fixed effects based on the pre-reform winner; column (3) further adds Census 2000 baseline characteristics interacted with the post indicator (log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor). All regressions are weighted by population in 2000. Robust standard errors clustered at the state level are reported in parentheses.

Endogeneity of realized expansion. A separate exclusion-related concern is reverse causality between political strength and realized enrollment. If incumbents who were electorally stronger before the reform were also more able or willing to expand enrollment afterward, direct estimates of the effect of realized expansion on vote share would be confounded. Table 5 regresses realized beneficiary expansion on measures of the incumbent’s 2008 electoral strength.

Table 5: Pre-Reform Electoral Strength and Subsequent Realized Expansion

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
Realized BF Expansion _{<i>m</i>} ($\hat{\kappa}_1$)	-0.001 (0.059)	-0.005 (0.044)	-0.011 (0.013)	-0.041 (0.023)	-0.077 (0.044)	-0.007 (0.014)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	2,105	2,105	2,105	2,105	2,105	2,105

Notes: The dependent variable is realized BF beneficiary expansion between 2008 and the 12-month average of 2009–2012, standardized. The explanatory variable is incumbent electoral strength in 2008, measured alternatively as the 2008-winner’s vote share (columns 1–3) or an indicator for the sitting incumbent winning in 2008 (columns 4–6). All columns restrict to the municipalities in the DiD-IV estimation sample. Columns (1) and (4) are bivariate; columns (2) and (5) add party fixed effects based on the 2008 winner; columns (3) and (6) further add baseline characteristics (Census 2000 log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula). Robust standard errors clustered at the state level are reported in parentheses. All regressions are weighted by population in 2000.

The estimates are small in magnitude and statistically indistinguishable from zero, so we do not find robust evidence that pre-reform electoral strength predicts subsequent *Bolsa Família* expansion. The 2SLS design remains appropriate on conceptual grounds. Reverse causality (where municipalities expand enrollment more aggressively where the expected political return is higher, as documented in [Brollo et al., 2020](#)) is a priori plausible even when it does not register empirically in this placebo test. The federal quota shock supplies exogenous variation that neutralizes this potential channel and is the reason we report 2SLS rather than OLS estimates.

4.3.3 Monotonicity

Assumption 3 (Weak Monotonicity). *For every municipality m , a positive quota shock cannot reduce realized expansion: $\Delta\text{Beneficiaries}_m(z') \geq \Delta\text{Beneficiaries}_m(z)$ whenever $z' > z$, where $\Delta\text{Beneficiaries}_m(z)$ is the potential inter-election change in beneficiaries that municipality m would realize under shock value z .*

The third assumption is monotonicity: a positive quota shock cannot reduce realized expansion in any municipality. As in any IV setting, this is an identifying restriction rather than a testable hypothesis, since we never observe both potential outcomes for the same unit. We follow standard practice in the continuous-treatment IV literature ([Imbens and Angrist, 1994](#); [Angrist and Imbens, 1995](#)) by combining an institutional

argument with empirical diagnostics consistent with weak monotonicity.

The institutional argument is direct. The 2009 reform raised the authorized number of beneficiary slots; it did not reduce them anywhere. A municipality that received a positive shock acquired headroom to enroll additional families if it chose to, while one that received a smaller adjustment lost no pre-existing capacity. There is no mechanical channel through which a higher federal authorization depresses enrollment. Defier behavior is in principle possible (a mayor could choose to scale back the registration apparatus precisely when the federal ceiling expanded), but we view this as empirically rather than institutionally implausible. The diagnostics below bound how much such behavior could distort the headline coefficient.

Three empirical diagnostics support this institutional reading. First, 67 percent of municipalities exhibit sign agreement between the inter-election change in beneficiaries and the quota shock, and these municipalities account for 98 percent of the IV-numerator mass $|\Delta D_m \cdot Z_m|$, which is the object that determines the 2SLS coefficient; the remaining 33 percent of sign-discordant municipalities are concentrated near zero on both axes (small expansion changes paired with small shocks) and contribute only 1.7 percent of the identifying mass. Second, a binned first-stage exercise tests whether the implementation chain runs in the same direction across the full support of the instrument: we sort municipalities into quintiles of Z_m and compute the conditional mean of ΔD_m within each bin. The conditional mean rises monotonically across the bulk of the distribution where the identifying variation concentrates, with the highest quintile (mean $Z_m \approx 1,446$ additional authorized slots) producing roughly six times the inter-election expansion of the next-highest quintile. Third, Section 5.5 reports the first-stage Kleibergen-Paap F -statistic across three pre-specified partitions (Northeast, the in-Frey saturation cell, and the partisan PT/non-PT cut) and finds it positive and well above the weak-instrument threshold in every subgroup that identifies the IV, with no subgroup exhibiting a wrong-signed coefficient. This is a stronger version of average monotonicity than the binned check, since the first stage runs in the same direction not only across quintiles of the instrument but across institutionally meaningful partitions of the sample.

Under the average-causal-response framework of Angrist and Imbens (1995) and Callaway et al. (2024), the headline 2SLS coefficient is a mass-weighted average of municipality-specific responses, with the weights determined by the sign-coherence between ΔD_m and Z_m documented above. Adapting the sensitivity bound of de Chaisemartin (2017) to this continuous-treatment setting, sign-discordant municipalities contribute negative weight to the pooled coefficient and carry 1.7 percent of the absolute

IV-numerator mass $|\Delta D_m \cdot Z_m|$. Even under the worst-case assumption that the sign-discordant municipalities have causal responses of magnitude equal and direction opposite to the sign-concordant ones, the implied sign-concordant ACR is bounded in $[+2.94, +3.03]$ percentage points, a 3.2 percent bandwidth around the headline of +2.94. The estimator is therefore essentially insensitive to monotonicity violations at any plausible magnitude.

4.3.4 The 2012 Quota Revision

Municipal authorization margins were revised again in 2012, shortly before the election.¹⁹ This revision was considerably smaller than the 2009 expansion but may still have affected 2012 electoral outcomes. We treat the 2009 revision as the primary source of identifying variation because it is the first reform after the 2008 election and has the full inter-election period to affect registration efforts and voter perceptions. The 2012 update is both smaller in magnitude and much closer to the election. Appendix C.3 reports supplementary specifications that extend the measurement window for the endogenous variable through 2012 and an updated version of the instrument using information from the 2010 Census.

5 Main Results

This section reports the main empirical results. We proceed in four steps. First, we establish that the 2009 quota reform generated substantial variation in realized program expansion (first stage). Second, we show that this exogenous variation in authorized quotas translates into better incumbent electoral performance: the intention-to-treat (ITT) effect of the quota shock on the ballot box, which is algebraically the reduced form. Third, we combine these two moments to recover the causal effect of instrumented beneficiary expansion on incumbent vote share and reelection (two-stage least squares). Fourth, we subject these estimates to a battery of robustness and heterogeneity checks. Throughout, both the year-varying treatment ($\text{Beneficiaries}_{mt}$) and the time-invariant instrument (ΔQuota_m) are standardized to mean zero and unit standard deviation within the estimation sample, so coefficients read as the response to a one-standard-deviation move in the relevant variable.

Each table presents six columns: three specifications for each outcome. Columns (1)

¹⁹Following the release of the 2010 Census, the federal government updated municipal poverty estimates used to guide program expansion in 2012. See the MDS *Informe n° 324*.

and (4) include only year and municipality fixed effects. Columns (2) and (5) add incumbent-party fixed effects based on the pre-reform winner’s party, capturing partisan baselines. Columns (3) and (6), our preferred specifications, additionally absorb five pre-treatment baseline characteristics interacted with the post-reform indicator: log population, the share of poor households, household income per capita, urban share, and the 2000 to 2006 population growth factor that enters the counterfactual quota formula (all from the 2000 Census).²⁰ All regressions are weighted by municipal population in 2000 and cluster standard errors at the state level.

5.1 First Stage: Authorization and Coverage

Table 6 reports the first-stage relationship between the quota shock and realized beneficiary expansion.

Table 6: First-Stage Estimates

	Realized Expansion in 2009		
	(1)	(2)	(3)
$\Delta \text{Quota}_m (\pi_1)$	0.199 (0.003)	0.197 (0.003)	0.205 (0.006)
Party Fixed Effects		✓	✓
Baseline Characteristics			✓
Observations	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses.

²⁰The incumbent’s 2008 vote share is *excluded* from the preferred specification. Absorbing the lagged outcome through its interaction with *post* would control for baseline incumbent strength, but combining unit fixed effects with a lagged dependent variable is known to introduce bias in short panels (Nickell, 1981; Imai and Kim, 2019). Our identification flows from the instrument, not from OLS on the lag, so the concern applies to nuisance absorption rather than the causal estimate; we keep the lagged vote share out of the main spec to be conservative. Column (2) of Table 9 adds it back as a sensitivity check.

In the preferred specification, which adds incumbent-party fixed effects and the baseline characteristics interacted with post, a one-standard-deviation increase in the quota shock raises year-varying standardized beneficiary enrollment by 0.205 standard deviations (SE 0.006). The estimate is virtually unchanged across specifications: 0.199 in Column (1), 0.197 in Column (2), and 0.205 in Column (3). The stability is informative. Because the 2009 quota shock is time-invariant and enters interacted with Post_t , the first-stage coefficient identifies the response of the muni-level change in beneficiaries to the shock; this object is structurally orthogonal to the time-invariant baseline characteristics that enter the regression as $\text{Census}_m^{2000} \times \text{Post}_t$, so adding those controls primarily affects the residual variance and the standard error. The Kleibergen-Paap F -statistic for the preferred column is 1,257, well above any conventional threshold for instrument strength.

The pass-through is economically substantial: one standard deviation of the shock corresponds to approximately 1,447 additional authorized beneficiaries, and the implied response in realized enrollment is approximately 1,015 additional families (0.205 panel-SD on the realized scale of 4,949 families per SD). Roughly 70 percent of the additional federal authorization translates into realized municipal coverage.

5.2 Reduced Form: Authorization and Electoral Returns

Table 7 reports the intention-to-treat (ITT) effect of the 2009 quota reform on incumbent electoral performance. The ITT regresses the electoral outcome directly on the exogenous quota shock, bypassing the second-stage instrumentation entirely. It is the most transparent test of whether being authorized to expand the program translates into electoral gains, and is causal under instrument exogeneity alone, since it does not require the exclusion restriction that the quota shock affects votes only through the realized beneficiary expansion channel. Interpreted as an ITT, it is the effect of the federal *authorization* rather than the effect of the additional families actually enrolled.

In the preferred specification, a one-standard-deviation increase in the quota shock raises the incumbent's vote share by 0.60 percentage points (SE 0.22) and the probability of winning by 0.84 percentage points (SE 0.27). Both estimates are significant at the one percent level. The ITT effect on vote share is stable across specifications, moving from 0.49 in Column (1) to 0.64 in Column (2) and 0.60 in Column (3); the small step from Column (2) to Column (3) is consistent with limited selection on observables as discussed earlier. For winning, Columns (4) and (5) yield raw coefficients of 1.31 and 1.78, attenuating to 0.84 once the baseline characteristics enter, consistent

Table 7: Reduced-Form Estimates

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{Quota}_m(\rho_1)$	0.494 (0.049)	0.637 (0.105)	0.603 (0.218)	1.305 (0.136)	1.784 (0.216)	0.841 (0.266)
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses.

with pre-existing electoral heterogeneity inflating the unconditional ITT before the preferred specification nets it out.

5.3 Second Stage: Coverage and Electoral Returns

Table 8 reports two-stage least squares estimates of the effect of instrumented beneficiary expansion on incumbent electoral performance. Under the first-difference reduction in equation (21), the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in vote share.

In the preferred specification, a one-standard-deviation increase in instrumented beneficiary expansion raises the incumbent’s vote share by 2.94 percentage points (SE 1.10) and the probability of reelection by 4.10 percentage points (SE 1.36). For vote share, the Anderson-Rubin 95 percent confidence set is [0.75, 5.25]; for winning, the set is [1.40, 7.00]. Both sets exclude zero under arbitrary instrument weakness.

The standardized coefficient is informative for hypothesis testing but obscures the natural-units interpretation. Two complementary scales help. First, since the cross-municipal standard deviation of the inter-election *change* in beneficiaries is approxi-

Table 8: Two-Stage Least Squares Estimates

	Vote Share			Winning		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \widehat{\text{Beneficiaries}}_m (\beta_1)$	2.476 (0.244)	3.228 (0.527)	2.935 (1.105)	6.541 (0.739)	9.035 (1.162)	4.096 (1.360)
Kleibergen-Paap F	3705.20	3624.90	1256.91	3705.20	3624.90	1256.91
AR 95% CI	[2.00, 2.95]	[2.15, 4.30]	[0.75, 5.25]	[5.05, 8.05]	[6.70, 11.45]	[1.40, 7.00]
Party Fixed Effects		✓	✓		✓	✓
Baseline Characteristics			✓			✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: Each coefficient is estimated on the municipality-year panel of 2008 and 2012 elections for the 2,105 municipalities where the 2008 incumbent ran for reelection in 2012. The endogenous variable is the year-varying twelve-month moving average of *Bolsa Família* beneficiaries; the instrument is the time-invariant 2009 quota shock interacted with the post-2009 indicator. Under two-period two-way fixed effects, the coefficient identifies the marginal effect of the muni-level inter-election change in beneficiaries on the corresponding change in the outcome (first-difference identification). All specifications include year and municipality fixed effects. “Party Fixed Effects” are based on the pre-reform winner’s party. “Baseline Characteristics” bundles five Census 2000 controls, each interacted with the post-reform indicator: log population, share of poor, household income per capita, urban share, and the 2000–2006 population growth factor that enters the counterfactual quota formula. All regressions are weighted by municipal population in 2000. Robust standard errors clustered at the state level (26 clusters) are reported in parentheses. The Kleibergen-Paap F -statistic is the squared t -ratio of the quota shock in the first-stage regression with cluster-robust standard errors. Anderson-Rubin 95% confidence sets are robust to arbitrary instrument weakness. An alternative ANCOVA-style specification, with the endogenous variable interacted with the post indicator, is reported in Appendix C.9.

mately 947 families, a typical inter-election expansion of one standard deviation raises the incumbent’s vote share by $2.94 \times 947/4,949 \approx 0.56$ percentage points. Second, in terms of the raw enrollment count, an additional 1,000 beneficiaries between the 2008 and 2012 elections raise the incumbent’s vote share by approximately 0.59 percentage points ($2.94/4.949$). The two scales describe the same coefficient under the panel-SD standardization adopted throughout the paper.

The 2SLS estimates are materially larger than their OLS counterparts: the OLS coefficient on the same standardized endogenous variable is 1.00 percentage points on vote share (against the 2.94 percentage points 2SLS estimate). A Wu-Hausman test rejects exogeneity for vote share at the one percent level, confirming the need for the IV strategy; for the binary winning indicator the test does not reject at conventional

levels.²¹

The preferred estimate of 2.94 percentage points is large in absolute terms. It is approximately equal to the typical inter-election decline in incumbent support of 3.00 percentage points in our sample, consistent with the magnitude of incumbency disadvantage that [Klašnja and Titiunik \(2017\)](#) document for Brazilian municipal elections using a regression-discontinuity design on close races. An exogenously induced one-SD expansion in beneficiary coverage therefore essentially cancels the average incumbency disadvantage between elections. The effect is large enough to be decisive in the typical close race: 26 percent of races in the analysis sample were decided by fewer than 5 percentage points in 2012, and 11 percent by fewer than 2 percentage points.

The estimate identifies an average causal response in the sense of [Angrist and Imbens \(1995\)](#): the response is a weighted average across municipalities, with weights concentrated where the 2009 quota shock generated meaningful enrollment changes. We document this concentration of identifying mass directly in [Appendix C](#), where we split the sample by the sign and the median of ΔQuota_m ; the response is identified primarily in the positive-shock regime, where the first stage is strong, while the negative-shock regime contributes negligible identification.

5.4 Robustness

The 2SLS estimate is robust to the standard set of perturbations of the modelling environment. The reduced-form effect retains significance under wild cluster bootstrap ([Rademacher and Webb](#)), under alternative clustering at the mesoregion and microregion levels, and under Conley standard errors at distance bandwidths of 100 and 200 kilometers. The two-stage least-squares estimate is significant under each of these procedures as well.

[Table 9](#) presents the core sensitivity analysis across six specifications. Column (1) reproduces the preferred estimate ($\hat{\beta}_1 = 2.94$). Column (2) adds the 2008 incumbent's vote share interacted with post to the baseline-characteristics bundle, addressing the lagged-dependent-variable concern discussed earlier; the point estimate is 2.52 and remains significant, confirming that the headline result is not an artifact of excluding the

²¹The Wu-Hausman test has lower power on a binary outcome with limited tail variation than on a continuous outcome, so failure to reject for winning is partly a power result. The IV and OLS coefficients for winning have the same sign and comparable magnitudes, and [Appendix C.4](#) reports both in parallel. We report 2SLS for both outcomes for internal consistency with the framework, in which both outcomes inherit the same underlying causal response. The IV-OLS gap on vote share is interpretable as evidence of negative political selection: mayors facing stronger electoral pressure expand coverage more aggressively, generating a downward bias in OLS that the federal authorization shock removes.

Table 9: Robustness Checks: Vote Share

	Main	Lag VS 2008	Ext. Window	Winsorized	Alt. Denominator	IPW
<i>Panel A: First Stage</i>						
$\Delta \text{Quota}_m (\pi_1)$	0.205 (0.006)	0.205 (0.006)	0.382 (0.014)	0.205 (0.006)	0.205 (0.006)	0.207 (0.005)
<i>Panel B: Reduced Form</i>						
$\Delta \text{Quota}_m (\rho_1)$	0.603 (0.218)	0.517 (0.195)	0.603 (0.218)	0.547 (0.207)	0.566 (0.188)	0.580 (0.228)
<i>Panel C: Two-Stage Least Squares</i>						
$\widehat{\Delta \text{Beneficiaries}}_m (\beta_1)$	2.935 (1.105)	2.524 (0.994)	1.577 (0.566)	2.662 (1.042)	2.756 (0.958)	2.804 (1.138)
Kleibergen-Paap F	1256.91	1243.27	800.79	1256.91	1256.91	1977.00
AR 95% CI	[0.75, 5.25]	[0.55, 4.65]	[0.45, 2.70]	[0.60, 4.85]	[0.85, 4.80]	[0.55, 5.20]
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All columns report the preferred specification: party fixed effects + baseline characteristics $\times$ post, population-2000 weights. The outcome is incumbent vote share except where noted. Column (2) adds the 2008 incumbent vote share $\times$ post to the baseline-characteristics bundle, testing sensitivity to including a lagged dependent variable (Nickell, 1981; Imai and Kim, 2019); the main specification omits this term. Column (3) uses the 12-month moving average of realized beneficiaries through December 2012 as the endogenous variable. Column (4) winsorizes the outcome at the 1st and 99th percentiles. Column (5) uses total votes cast (including blank and null) as the denominator. Column (6) weights by the inverse probability of selection into the rerunning sample. Anderson-Rubin 95% confidence sets are robust to arbitrary instrument weakness. Robust standard errors clustered at the state level in parentheses.

lagged outcome. Columns (3) to (6) perturb the treatment, outcome, or sample while holding the instrument fixed: extending the treatment-window to December 2012 (Column 3) yields $\hat{\beta}_1 = 1.58$ (still significant under the AR confidence set of [0.45, 2.70]).

Column 3 redefines the endogenous variable as the change in the 12-month beneficiary count between the close of 2008 and the close of 2012, measuring the expansion as near to the ballot as the data allow; it yields $\hat{\beta}_1 = 1.58$, still significant under the AR confidence set [0.45, 2.70]. The attenuation reflects the 2012 quota revision, which used the 2010 Census to recalibrate municipal poverty estimates and reduced authorized quotas in municipalities whose 2010 poverty figures fell below the 2009 projections, partially reversing the 2009 over-allocation in those places. A window running through December 2012 therefore averages the 2009 expansion with this partial 2012 reversal. We prefer the headline window because it aligns the treatment with the period the federal government actually assigned: the 2009 reform instructed municipalities to fill their newly authorized quotas during 2009, so the 2009 enrollment change is the policy-relevant treatment, while the 1.58 estimate is best read as a window-sensitivity diagnostic that integrates over a later, partly offsetting reform.

5.5 Heterogeneity

We close the main results with a heterogeneity exercise along three sample partitions: Northeast region, the population-by-HDI cell that anchors Frey (2019), and the 2008-elected incumbent’s party. Table 10 reports the saturated dual-IV decomposition for vote share. In each partition, the instrument and the endogenous variable are interacted with a binary indicator and its complement, yielding two coefficients per equation from a single full-sample regression. The decomposition tests the augmented model’s central prediction that the first stage attenuates on partitions aligned with the structural margins the model and remains comparable on partitions orthogonal to them.

Table 10: Heterogeneity by Subgroup: Vote Share (Dual-IV)

	Northeast			Frey 2019			PT incumbent		
	NE	non-NE	Δ	in Frey	out Frey	Δ	PT	non-PT	Δ
<i>Panel A: First Stage</i>									
$\Delta\text{Quota}_m(\pi_1)$	0.270 (0.031)	0.212 (0.003)	0.058 (0.032)	0.029 (0.011)	0.205 (0.006)	-0.176 (0.013)	0.169 (0.014)	0.203 (0.005)	-0.034 (0.015)
<i>Panel B: Reduced Form</i>									
$\Delta\text{Quota}_m(\rho_1)$	-0.324 (0.533)	0.611 (0.220)	-0.934 (0.373)	-2.257 (2.149)	0.600 (0.217)	-2.858 (2.108)	1.051 (0.343)	0.580 (0.229)	0.471 (0.212)
<i>Panel C: Two-Stage Least Squares</i>									
$\Delta\widehat{\text{Beneficiaries}}_m(\beta_1)$	-1.531 (1.860)	2.836 (1.114)	-4.367 (1.149)	-98.872 (74.189)	3.040 (1.103)	-101.911 (74.086)	6.162 (2.493)	2.725 (1.170)	3.437 (1.671)
Kleibergen-Paap F	74.08	3827.53		6.65	1214.32		146.97	1898.01	
AR 95% CI	[-5.15, 3.10]	[0.75, 5.15]		[-15.00, 25.00]	[0.75, 5.25]		[1.80, 12.30]	[0.55, 5.25]	
Party Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from a single full-sample dual-IV regression per heterogeneity dimension: the instrument and the endogenous variable are each interacted with a binary subgroup indicator and its complement, producing two coefficients per equation from one regression. Columns (1)–(3) split on Northeast region (AL, BA, CE, MA, PB, PE, PI, RN, SE) versus other regions. Columns (4)–(6) split on the Frey (2019) sample restriction (population in 2000 below 30,000 and Atlas IDHM 2010 below 0.7), with column (4) inside Frey’s sample and column (5) outside. Columns (7)–(9) split on whether the 2008-elected mayor’s party (winner_party_2008) — the sitting incumbent during the 2009–2012 expansion window — is the Workers’ Party (PT) versus any other party. Δ is the Wald test of equality of the two coefficients reported as the difference with its standard error. Specification matches the headline tables: 90/120 quota shock instrument, year-varying realized beneficiaries (12-month MA, standardized) as the endogenous variable, party fixed effects (pre-reform winner) plus baseline characteristics $\times$ post (Census 2000 log population, share of poor, household income per capita, urban share, and 2000–2006 population growth factor), municipality and year fixed effects, weighted by 2000 population, with cluster-robust standard errors at the state level. Anderson-Rubin 95% confidence sets are computed by grid inversion holding the other group’s coefficient at its dual-IV estimate; they are robust to weakness of the corresponding sub-instrument.

The Northeast partition (columns 1–3) leaves the first stage intact in both subsam-

ples: the instrument is strong inside and outside the region, comfortably above the Stock-Yogo threshold in each. The electoral response, by contrast, concentrates outside the Northeast, where the 2SLS estimate tracks the pooled headline; inside the region it is statistically indistinguishable from zero. The headline is therefore identified primarily from non-Northeastern compliers. What matters for the framework is that the regional split does not switch the first stage on or off, so region is not one of the structural margins the augmented model predicts should govern identification.

The Frey partition (columns 4–6) is sharper and more telling. Inside the cell that anchors Frey (2019)’s identification the first stage collapses below the conventional weak-instrument threshold, the corresponding 2SLS estimate is uninterpretable noise, and identification comes entirely from outside the cell, where the response again matches the pooled headline. The collapse is mechanical rather than structural: Frey’s anchor cell coincides with the small-population, medium-HDI subsample in which *Bolsa Família* had already reached high saturation by the late 2000s, so the eligible pool was largely exhausted and the federal authorization had no extensive margin to push into. With the pre-existing utilization rate B_m/Quota_m in this cell already close to one by 2008, a positive shock cannot generate realized expansion where the room for it is gone. Appendix D.2 confirms this collapse is robust to using direct saturation cuts (bottom quartile, median, mean, and top quartile of pre-reform beneficiaries per 100 inhabitants) as the binary partition rather than Frey’s institutional indicator, and develops the substantive distinction between Frey’s identification and ours: Frey identifies the electoral return to a federally-controlled enrollment channel under saturation, while we identify the return to a locally-controlled channel where there is room left to grow.

The PT partition (columns 7–9) is qualitatively different from the other two and is the most substantively interesting heterogeneity in the table. Both subgroups identify the effect at both stages, with a strong first stage on each side and 2SLS coefficients that exclude zero. The contrast is in magnitude: the return where the incumbent is from the Workers’ Party, $\hat{\beta}_1$ of about 6.2 percentage points, is more than double the return elsewhere, and a Wald test rejects equality of the two at the five percent level. The asymmetry sits in the per-beneficiary return rather than in the first stage; if anything the conversion of authorization into realized enrollment is somewhat slower under PT mayors, yet each marginal beneficiary delivers more than twice the vote share. This is the shape the framework commits to in advance: the partisan-attribution extension of the voter-response block (Section 3.4, equation (16)) raises the per-beneficiary return by $\lambda_2^{\text{attrib}} \geq 0$ wherever the mayor’s party also operates the federal program, while

holding the first stage partisan-invariant because the material channel λ_1 is federally set. The partition corroborates the prediction on both margins.

The mechanism behind $\lambda_2^{\text{attrib}}$ is institutional. *Bolsa Família* was created in 2003 under the federal Lula administration and has been the Workers' Party's flagship anti-poverty policy ever since; the 2009 reform that supplies our identifying variation was implemented under a PT-controlled federal executive in Lula's second term, and the 2012 election that we study was held under the first PT presidency of Dilma Rousseff. A municipality with a PT mayor between 2008 and 2012 is therefore administering a program created and politically owned by the same party at the federal level, throughout a period of continuous PT control of the federal executive. Voter attribution under these conditions is unambiguous: the partisan chain of credit runs cleanly from the local registration apparatus to the federal program design, while the same expansion under a non-PT mayor splits credit between local administration and federal sponsorship. We read the asymmetry as the empirical signature of this attribution channel rather than as a separate structural margin, and as a substantive component of the paper's main message: centralized program design does not, on its own, depoliticize redistribution, and the absence of depoliticization is sharpest where the local administrator and the federal program creator share a partisan brand.

6 Possible Mechanisms

The main results establish that federally induced quota expansion generated electoral returns for incumbent mayors. This section uses the same framework to test the augmented model's structural predictions about which margins of capacity must be in place for the federal-to-local chain to operate. We organize the evidence around the two margins of Section 3.3: infrastructure (κ^I) and alignment (κ^A).

First, local welfare infrastructure operates as the binding margin the augmented model predicts: without a CRAS in place before the reform, the quota shock attenuates by roughly a factor of three in the first stage relative to CRAS municipalities, and the electoral coefficient is too imprecisely estimated to interpret.

Second, among CRAS municipalities, the alignment margin operates through manager substitution. We document a chain of evidence consistent with the stepping-stone interpretation of Colonnelli et al. (2020): the federal shock causes substitution of incumbent-CRAS managers, but does not cause new CRAS unit openings, and the substitutes hired into post-shock units run for elected office at higher rates than the

baseline without switching to the mayor’s party and without prior partisan-affiliation selection. Roughly three quarters of the manager changes our definition captures bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles within or across CRAS units, ruling out the reading that the turnover margin is a relabeling exercise.

We frame these findings as causally identifying the average causal response of incumbent vote share to realized beneficiary expansion within the CRAS-with-turnover subgroup. Three pieces of evidence support this reading. The within-subgroup first stage is essentially zero in the inactive cells (Anderson-Rubin sets unbounded; Kleibergen-Paap F at 1.3 in CRAS-no-turnover; the Wald test of equality across cells rejects decisively), so the inactive cells contribute zero compliers to the pooled IV. The Frisch-Waugh-Lovell decomposition concentrates the identifying covariance overwhelmingly on the CRAS-with-turnover cell. The classification of manager changes into external, internal-same-CRAS, and internal-other-CRAS hire types (75 percent external) rules out the alternative reading that the turnover indicator captures bureaucratic re-shuffles rather than fresh personnel inflows.²²

6.1 State Capacity as a Precondition

Of the 2,105 municipalities in the estimation sample, 1,415 (67.2 percent) had at least one CRAS in the 2008 *Censo SUAS*; the remaining 690 did not. The augmented model is unambiguous about the implication: the within-subgroup first stage should attenuate by roughly a factor of γ in $\{\kappa_m^I = 0\}$, where γ is the productivity floor of the implementation chain. The benchmark, by contrast, predicts smooth attenuation of comparable magnitude. The dual-IV decomposition that follows tests the two against each other.

Table 11 reports the specification that identifies the effect separately for CRAS municipalities and for municipalities without a CRAS. Both the instrument and the endogenous variable are interacted with $\mathbf{1}(\text{CRAS}_m^{2008})$ and with its complement $1 - \mathbf{1}(\text{CRAS}_m^{2008})$, producing two endogenous coefficients in a single regression.

The first stage and reduced form deliver the asymmetry the augmented model pre-

²²All mechanism specifications use the preferred instrument, the preferred endogenous variable, and the Column (3) control set of Section 5: incumbent-party fixed effects based on the pre-reform winner and baseline characteristics $\times$ post. Appendix D reports additional mechanism dimensions, Appendix E.2 reports a within-CRAS test of operational intensity proxies (*busca ativa* and *PAIF* tracking) that confirms the augmented model’s prediction that non-structural intensity margins should not discriminate first-stage strength, and Appendix C.10 reports a separate robustness check that confirms the headline is not confounded by the concurrent federal increase in per-family benefit values announced in March 2011.

Table 11: CRAS Presence as Necessary Condition

	Vote Share			Winning		
	has = 1	has = 0	Δ	has = 1	has = 0	Δ
First Stage (π_1)	0.208 (0.005)	0.075 (0.012)	0.133 (0.013)	0.208 (0.005)	0.075 (0.012)	0.133 (0.013)
Reduced Form (ρ_1)	0.611 (0.219)	-2.636 (3.072)	3.247 (2.983)	0.803 (0.273)	7.274 (10.210)	-6.471 (10.087)
Two Stage Least Squares (β_1)	2.947 (1.078)	-37.640 (44.880)	40.588 (44.429)	3.841 (1.380)	94.565 (143.600)	-90.724 (143.115)
Kleibergen-Paap F	1734.99	35.64		1734.99	35.64	
AR 95% CI	[0.80, 5.20]	[-15.00, 25.00]		[1.15, 6.75]	[-15.00, 25.00]	
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from the full-sample dual-IV regression: the instrument and the endogenous variable are each interacted with $\mathbf{1}(\text{CRAS}_m^{2008})$ and its complement, producing two coefficients per equation from a single regression. The “has = 1” column reports the coefficient on the CRAS-interaction; “has = 0” reports the coefficient on the no-CRAS interaction. Δ is the Wald test of $\beta_{\text{has}=1} = \beta_{\text{has}=0}$ reported as the difference with its standard error and $\chi^2(1)$ p -value. Under the preferred 90/120 instrument the first stage is identified in both subgroups (Kleibergen-Paap F above the Stock-Yogo 10%-bias threshold), so the 2SLS coefficient and Anderson-Rubin 95% confidence interval are reported for both “has = 1” and “has = 0”. All specifications include year and municipality fixed effects, party fixed effects based on the pre-reform winner (2004-winner in 2008; 2008-winner in 2012), and baseline characteristics $\times$ post: log population, Gini inequality, share of poor, household income per capita, share of formal workers, and urban share (all from Census 2000). Weights: municipal population in 2000. Cluster-robust standard errors at the state level in parentheses.

dicts. The CRAS subsample reproduces the pooled-sample first stage almost exactly, while the no-CRAS subsample retains a small but statistically identified pass-through that is roughly a factor of three weaker (a Wald test of equality across the two cells rejects). The CRAS-subgroup reduced-form responses on vote share and reelection are essentially indistinguishable from the headline; the no-CRAS responses are too imprecise to interpret and comfortably contain zero. The pattern is consistent with the broader finding in the *Bolsa Família* literature that implementation, not design, is where municipal governments shape distributive outcomes (Fried, 2012; Sugiyama and Hunter, 2013).

The 2SLS coefficients sharpen the asymmetry rather than add to it. The CRAS-subgroup point estimates and Anderson-Rubin confidence sets reproduce the pooled headline directly. The no-CRAS subgroup inherits the well-known weak-instrument inflation: the first-stage pass-through there is identifiably positive, but quantitatively too small to support a meaningful 2SLS denominator, and the AR set spans the en-

tire admissible range. The Frisch-Waugh-Lovell weights formalize this directly: in the saturated specification, $\omega_{\text{CRAS}} \approx 0.89$ and $\omega_{\text{no-CRAS}} \approx 0.11$, so nearly nine-tenths of the identifying covariance behind the headline 2SLS comes from the CRAS subgroup, with the no-CRAS contribution residual.

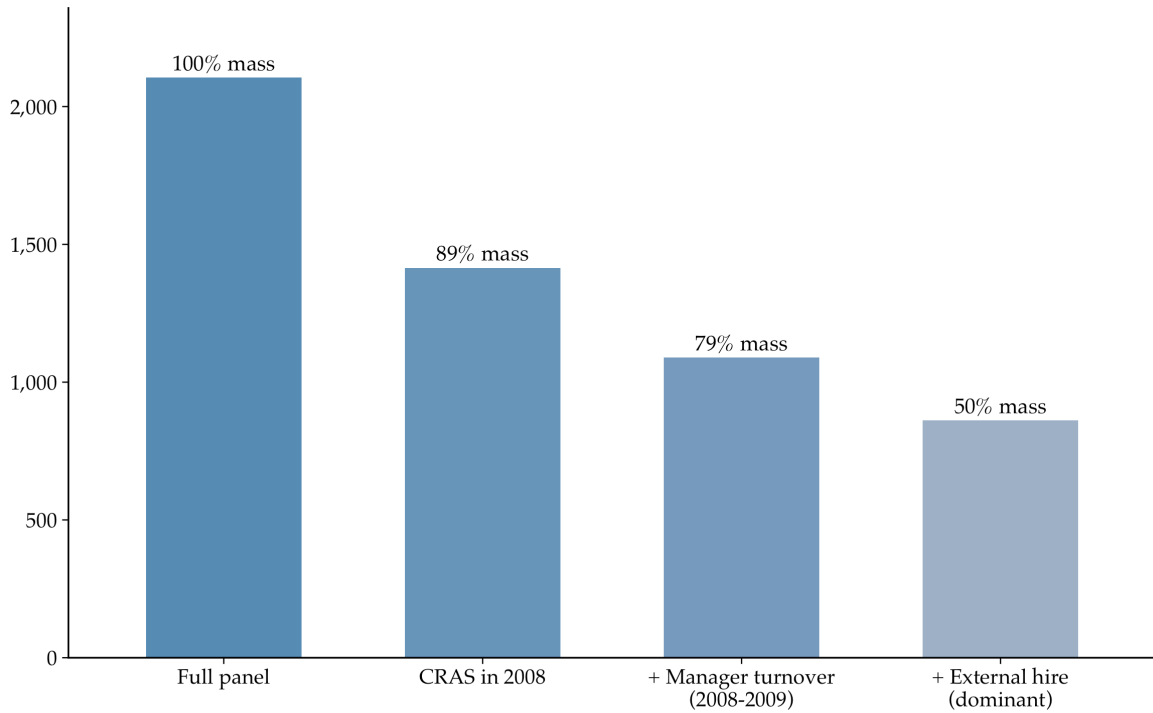
The substantive content of the table is therefore the asymmetry the augmented model anticipates. The productivity floor γ governing the inactive-subgroup attenuation is small enough that the no-CRAS first stage does not respond enough to the federal authorization to generate an interpretable second stage. The headline electoral return is the per-beneficiary effect for compliers in CRAS municipalities, where the implementation chain operates; it neither confirms nor rejects the existence of an electoral return in the no-CRAS subgroup, where the data are silent.

6.2 Manager Alignment as Electoral Stepping Stone

Figure 4 previews the mechanism cascade. From the full panel of 2,105 municipalities, restricting to pre-reform CRAS municipalities (1,415) retains 89 percent of the implicit Frisch-Waugh-Lovell identifying mass; further restricting to those where the incoming mayor replaced the *de facto* CRAS manager on taking office (1,089) retains 79 percent; and conditioning on the dominant hire type being external (861) retains 50 percent. The cascade visualizes the central empirical claim of this section: the headline effect is concentrated where the structural margins of Section 3.3 are jointly favorable, and the per-step shrinkage in muni count is sublinear in the mass retained, exactly the pattern the augmented model predicts.

The augmented model predicts that the alignment margin κ^A is the second binding constraint: the within-subgroup first stage should attenuate sharply in CRAS municipalities where the incoming mayor did not replace the front-line manager on taking office. Table 12 reports the corresponding decomposition, which estimates three coefficients in a single full-sample regression: no CRAS in 2008, CRAS where the manager stayed, and CRAS where the new mayor replaced the manager. We define the manager as the *de facto* frontline operator: the respondent of the unit-level *Censo SUAS* questionnaire, who is the person who actually fills the form and runs the unit day-to-day. The 2008 and 2009 *Censo SUAS* are annual snapshots recorded just before and just after the mayor elected in 2008 takes office, so a change in the recorded manager between the two waves identifies a replacement made by the incoming administration; the *de facto* definition admits an exact CPF match across waves, lets us classify changes as external hires versus internal transfers, and is the object the structural argument in Section 3.3

Figure 4: Cascade of the Analysis Sample and Implicit Identifying Mass



Note. Bar height is the municipality count in each cascade cell. The percentage above each bar, also encoded by the bar shading, is the share of identifying mass the cell retains: the Frisch-Waugh-Lovell share of the residualized, population-weighted variance of the standardized 2009 quota shock $\times$ post that lives in the cell, expressed as a fraction of the full-panel total. Each step adds one restriction, from the full panel of municipalities to those with a pre-reform *CRAS*, then to those where the mayor elected in 2008 replaced the *de facto* *CRAS* manager on taking office (measured between the 2008 and 2009 *Censo SUAS* waves), then to those whose dominant manager change was an external hire. “External hire (dominant)” is the subset of *CRAS*-with-turnover municipalities in which the majority of *de facto* manager changes brought in a successor who was not present in the 2008 *CRAS* structure anywhere.

maps onto.

The first stage in *CRAS* municipalities with manager turnover reproduces the pooled headline almost exactly, while the other two cells collapse below the conventional weak-instrument threshold. The joint Wald test of equality across the three coefficients rejects decisively, and the Frisch-Waugh-Lovell decomposition concentrates roughly $\omega_{\text{turn}} \approx 0.93$ of the identifying covariance on the third column. The Anderson-Rubin confidence sets reported within each cell confirm the asymmetry directly: in the *CRAS*-with-turnover cell, the AR 95% set on vote share is [1.45, 5.50], comfortably bracketing the headline 2.94; in the *CRAS*-without-turnover cell, the AR set is essentially unbounded, the appropriate inference output when the first stage is statistically indistinguishable from zero. The channel operates where the person who actually runs the unit changes.

Table 12: Heterogeneity by Manager Turnover

	Vote Share			Winning		
	non-CRAS	CRAS / non-Turn	CRAS / Turnover	non-CRAS	CRAS / non-Turn	CRAS / Turnover
First Stage (π_1)	0.075 (0.013)	0.035 (0.010)	0.206 (0.004)	0.075 (0.013)	0.035 (0.010)	0.206 (0.004)
Reduced Form (ρ_1)	-2.370 (2.999)	2.831 (0.945)	0.624 (0.203)	7.795 (10.072)	4.350 (2.355)	0.951 (0.227)
Two Stage Least Squares (β_1)	-26.893 (48.413)	75.166 (43.912)	3.443 (1.250)	111.985 (140.496)	114.629 (49.528)	5.217 (1.439)
Kleibergen-Paap F	35.58	11.73	2623.64	35.58	11.73	2623.64
AR 95% CI	[-15.00, +25.00]	[+12.60, +25.00]	[+1.45, +5.50]	[-15.00, +25.00]	[-15.00, +25.00]	[+2.60, +7.95]
Party Fixed Effects	✓	✓	✓	✓	✓	✓
Baseline Characteristics	✓	✓	✓	✓	✓	✓
Observations	4,210	4,210	4,210	4,210	4,210	4,210

Notes: All estimates are obtained from a single full-sample triple-IV regression per outcome. The instrument and the endogenous variable are each interacted with three mutually exclusive indicators, ordered in the table from left to right: non-CRAS (municipalities with no CRAS in the 2008 *Censo SUAS*); CRAS / non-Turn (municipalities with a CRAS in 2008 but no manager change between the 2008 and 2009 waves); and CRAS / Turnover (municipalities with a CRAS in 2008 and a manager change between the 2008 and 2009 waves). The manager is defined as the *de facto* frontline operator: the *cpf_agente* respondent of the unit-level *Censo SUAS* questionnaire, the person who actually fills the form and operates the unit day-to-day, who is not necessarily the formal *Coordenador*. All specifications include year and municipality fixed effects, party fixed effects based on the pre-reform winner (2004-winner in 2008; 2008-winner in 2012), and baseline characteristics $\times$ post: log population, Gini inequality, share of poor, household income per capita, share of formal workers, and urban share (all from Census 2000). Weights: municipal population in 2000. Cluster-robust standard errors at the state level in parentheses. Anderson-Rubin 95% confidence intervals are reported for every cell and are valid under arbitrary instrument weakness; they are the appropriate inference object in the no-CRAS and no-turnover cells, where the Kleibergen-Paap F -statistic falls below the strong-instrument threshold.

The hire-type composition of those changes corroborates the reading. Across 3,357 manager changes in 2,583 municipalities, 75.5 percent involve a successor who was not present in the CRAS structure anywhere in 2008, 14.7 percent involve an internal transfer from another role at the same CRAS, and 9.8 percent involve a transfer from a different CRAS in the same municipality. The dominant case is fresh personnel, not internal reshuffles. Appendix E.5 reports the corresponding heterogeneity table. The within-CRAS-with-turnover return is identified primarily across the external-hire subsample, which carries the bulk of the identifying mass and yields a 2SLS point estimate close to the pooled headline; the internal-same-CRAS cell delivers a comparable first stage and a similar-magnitude vote-share return on a smaller subsample; the internal-other-CRAS cell is too small to generate a within-cell first stage that supports a precisely-estimated 2SLS. The heterogeneity table does not therefore sharply discriminate “external ability hire” from “internal-same-CRAS re-deployment”, and we read it as additional support for the substantive claim that the turnover indicator captures genuine personnel inflows rather than as a discriminating test between hire-type sub-

types.

6.2.1 What Does the Alignment Channel Actually Look Like?

The decomposition tells us that manager turnover is the binding margin within CRAS, but it leaves open what the substitute does once installed. The simplest reading is that incoming managers are partisan loyalists, but that account is at odds with the data.

Table 13: Manager Trajectory

	Parameter	N
<i>Panel A. Does the shock cause manager substitution?</i>		
CRAS units that persisted in 2008 and 2009	0.556 (0.243)	1,341
<i>Panel B. Does the shock select substitutes by prior partisan affiliation?</i>		
Any party affiliation pre-2009	0.826 (0.530)	1,099
<i>Panel C. Does the shock raise the rate at which substitutes run for office?</i>		
Probability of running for office (2010–2016)	0.450 (0.144)	1,099
<i>Panel D. Does the shock align substitutes with the mayor's coalition?</i>		
Joined mayor's party (2009–2012)	0.043 (0.045)	1,099
Joined mayor's coalition (2009–2012)	0.102 (0.051)	1,099
Joined any party (2009–2012)	-0.007 (0.109)	1,099

Each row reports a separate cross-sectional regression of the row outcome on the standardized 2009 quota shock plus Census-2000 controls (log population, share in extreme poverty, household income per capita, urban share, and 2000–2006 population growth), with state fixed effects and population weights. Standard errors (in parentheses) are clustered at the state level. The sample restricts to municipalities in the main regression sample with at least one incumbent CRAS unit observed in both the 2008 and 2009 *Censo SUAS* waves, with at least one substitution event (Panels B–D); Panel A uses the broader sample of municipalities with at least one incumbent unit. “Substitute” is a CPF observed in 2009 at an incumbent unit where the 2008 manager had a different CPF. The mayor's coalition is the set of parties listed in `winner_coalition_2008`. All coefficients and standard errors are in percentage points (multiplied by 100). The placebo for new CRAS unit openings is reported in Appendix E.1.

Does the shock move people, or does it move infrastructure? Panel A of Table 13 reports the answer to the first reader question. In the subsample of municipalities with at least

one incumbent CRAS unit observed in both the 2008 and 2009 *Censo SUAS* waves, the annual snapshots that bracket the change of mayor, a one-standard-deviation increase in the federal shock raises the probability that the unit experiences a manager substitution by about half a percentage point on a base of 82 percent. The placebo for new CRAS unit openings, reported in Appendix E.1, is a clean null. The shock moves managers, not buildings.

Was the substitute already political? Panel B asks whether the substitute was already a partisan operator before the 2009 reform. The shock raises the share of substitutes with an open pre-2009 party affiliation by only 0.83 percentage points, an effect that is not distinguishable from zero at conventional levels. The descriptive panel of Table 14 reinforces the result: stayers, outgoing managers, and incoming managers have similar pre-2009 affiliation rates, between 16.5 and 19.6 percent, and the gap between incoming and outgoing is not statistically significant. The shock did not select substitutes by prior partisan history.

Table 14: Substitute Manager Characteristics

<i>Panel A. Comparison across managerial roles (incumbent units, 2008–2009)</i>				
	# CPFs	% CPFs	Filiated	Ran
Stayer (same CPF 2008–2009)	269	91.5%	18.4%	5.8%
Outgoing (replaced in 2009)	1,225	89.7%	19.6%	6.5%
Incoming (substitute hired 2009)	1,216	89.0%	16.5%	9.5%
<i>Panel B. Offices sought by incoming managers who ran (TSE 2010–2016)</i>				
	Level	# CPFs	% runners	% substitutes
City council (<i>vereador</i>)	Municipal	99	81.8%	8.14%
Vice mayor	Municipal	18	14.9%	1.48%
Mayor (<i>prefeito</i>)	Municipal	10	8.3%	0.82%
State deputy	State	2	1.7%	0.16%
Any office		121	100.0%	9.95%

Panel A: Each row aggregates manager-CPFs at incumbent CRAS units observed in both the 2008 and 2009 *Censo SUAS* waves. *Stayer* is the same CPF in both years; *Outgoing* is the 2008 manager replaced in 2009; *Incoming* is the 2009 substitute. # CPFs counts distinct managers within the role; % CPFs reports the share of within-role observations that map to a distinct manager (i.e., one minus the duplicate rate from managers serving multiple units in the same municipality). *Filiated* is the share of CPFs with an open party affiliation as of 1 January 2009. *Ran* is the share of CPFs that appear as a candidate for any elected post in the TSE 2010–2016 candidate database. *Panel B:* For substitutes who ran for office, the breakdown by office category. CPFs running for multiple offices across 2010–2016 are counted once per office category, so row totals exceed the unique runner count of 121.

What does the substitute do after taking the post? Panel C of Table 13 reports the causal

evidence: among the substitutes installed in 2009, the probability of running for elected office between 2010 and 2016 rises with the size of the shock, and the response is statistically distinguishable from zero. Panel A of Table 14 reports a complementary descriptive contrast. The running rate of incoming managers, 9.5 percent, is nearly twice the rate among stayers, 5.8 percent; the gap survives when restricted to managers in the same units. The same contrast is absent between outgoing managers and stayers, so being a CRAS manager does not by itself raise the running rate; specifically, being the post-2008 entrant does. Panel B of Table 14 shows that the post-bureaucracy career path is overwhelmingly local: 99 of the 121 unique substitutes who ran did so for *vereador*, with another 18 also running for vice mayor, 10 for mayor, and 2 for state deputy. The office counts are not mutually exclusive because some substitutes ran for multiple offices across the 2010–2016 cycles, so the office-by-office sum exceeds the unique runner count of 121.

Does the substitute align with the local mayor? Panel D of Table 13 answers in two parts. The substitute does not switch to the mayor’s exact party at higher rates after the shock, and is not more likely to join any party in general. The substitute is, however, more likely to file a new affiliation with one of the parties in the mayor’s 2008 electoral coalition. The alignment that operates is broad-coalition, not narrow-party brand: the substitute is positioning herself within the local coalition that just won the mayoral race, not pledging fealty to a single party.

The composite picture is the stepping-stone channel of Colonnelli et al. (2020). The shock makes the CRAS managerial position more attractive as a launching pad for an electoral career: it expands the operational footprint of the office, raises the visibility of the person who runs it, and broadens that person’s procedural reach into local households. The substitutes the shock attracts are not selected on prior partisanship, but they treat the post differently than the inherited managers do, both in subsequent candidacy rates and in coalition affiliation. That difference is what activates the alignment margin κ^A in the model: the multiplier on discretionary effort a_m rises, and the federal authorization translates more strongly into realized expansion.

What the data cannot adjudicate is the mayor’s selection criterion: whether the substitute was hired on capacity grounds, on grounds of political alignment with the incoming coalition, or on a combination of the two. What is unambiguous is what the substitute does with the post once installed. The visibility of the expanded CRAS office is converted into a local electoral candidacy at higher rates than for stayers, regardless of why the mayor originally appointed her. Under either selection story (competent hire later activated by visibility, or politically ambitious hire pre-selected for the launch-

ing pad), the structural implication is the same: realized expansion e_m rises, the voter signal s_{im} shifts upward, the posterior $\Pr(\theta_m = 1 \mid s_{im})$ tracks it, and the inference channel $\lambda_2 e_m$ delivers the additional vote share that κ^A activates. The stepping-stone label refers to this downstream behavior of the substitute, not to a presumption about the mayor's selection process.

7 Conclusion

This paper asks whether local politicians can extract electoral returns from implementing a federally designed cash transfer program. We exploit the 2009 reform that revised the formula for allocating *Bolsa Família* beneficiary quotas across Brazilian municipalities, using the federally-induced authorization shock to instrument realized program expansion, and estimate its effect on the sitting mayor's performance in the 2012 municipal elections. A simple framework organizes the analysis: the federal-to-local implementation chain is multiplicative in physical infrastructure (κ^I , the pre-reform CRAS) and front-line bureaucratic alignment (κ^A , manager turnover). The framework yields a set of testable predictions about where the federal authorization translates into realized expansion, and where that expansion translates into electoral returns. Realized expansion is also bounded mechanically by the eligible-pool margin a municipality faces in 2008: where local saturation is high, there is little room for the shock to operate regardless of capacity.

A one-standard-deviation increase in instrumented beneficiary expansion, approximately 4,949 additional families in the estimation sample, raises the incumbent's vote share by 2.94 percentage points and the probability of reelection by 4.10 percentage points. In natural units, an additional 1,000 beneficiaries between elections corresponds to roughly 0.59 percentage points on vote share and 0.83 percentage points on reelection. The estimates are robust to alternative instrument thresholds, alternative measurement windows for the endogenous variable, alternative outcome definitions, alternative weights, and inverse-probability weighting for selection into the rerunning sample. Anderson-Rubin confidence sets, which remain valid under arbitrary instrument weakness and small cluster counts, exclude zero for both outcomes.

The mechanism evidence locates the headline along the structural margins the framework anticipates. Pre-existing local welfare infrastructure operates as the binding precondition: in municipalities without a CRAS at the time of the reform, the within-subgroup first stage attenuates by roughly a factor of three relative to CRAS municipalities and the corresponding electoral coefficient is too imprecise to interpret.

Within CRAS municipalities, the alignment margin operates through manager substitution: the federal shock causes turnover of the *de facto* front-line manager without causing new unit openings, and three quarters of the manager changes captured by our definition bring in individuals new to the municipal welfare bureaucracy rather than internal reshuffles. The substitutes hired into post-shock units run for elected office at higher rates than the baseline while aligning with the mayor’s electoral coalition rather than switching to the mayor’s exact party or being pre-selected on prior partisan affiliation. The pattern is most easily reconciled with the downstream-behavior half of the stepping-stone channel documented by [Colonnelli et al. \(2020\)](#), not their loyalist-selection half: the federal shock raises the visibility of the CRAS managerial post and makes it more attractive as a launching pad for a local electoral career regardless of the mayor’s selection criterion, an effort margin in which the structural alignment indicator κ^A becomes active. The asymmetry in returns extends to political alignment: per-beneficiary electoral returns are roughly 2.3 times larger under PT-aligned mayors than under non-PT mayors (6.16 versus 2.73 percentage points; Wald $p = 0.040$), an empirical signature of partisan attribution that the centralized-design literature does not address.

These results contribute to four literatures. The political-economy literature on cash transfers documents that incumbents reap electoral returns from program expansion in settings where local discretion is deliberately minimized ([Manacorda et al., 2011](#); [De La O, 2013](#)); we complement that work by studying the opposite institutional configuration, in which the program is centrally designed but the registration pipeline is locally operated. The comparison with [Frey \(2019\)](#), developed in Section 1, makes the sign contrast informative: the two designs identify electoral returns on different sets of municipalities and on different margins of local effort.

The political-accountability literature on shared governance ([Przeworski et al., 1999](#); [Besley, 2006](#); [Ferraz and Finan, 2008](#)) centers on the question of whether voters correctly attribute responsibility across levels of government; our setting differs in that the relevant information concerns positive delivery rather than malfeasance, but it operates through the same retrospective-inference logic and shows that local incumbents can be rewarded for federally-funded programs where local effort remains salient.

The state-capacity literature ([Finan et al., 2017](#); [Dal Bó et al., 2013](#)) emphasizes the personnel and infrastructure through which programs are delivered; our findings identify a margin prior to the quality of bureaucrats, the *existence* of the administrative apparatus, and a complementary margin within the apparatus, the alignment of the front-line manager with the local political principal, in the sense of [Akhtari et al. \(2022\)](#) and

Colonnelli et al. (2020).

Two implications follow. First, centralized program design does not, on its own, depoliticize redistribution: even when eligibility rules, benefit levels, and payment channels are set nationally, local discretion over the registration pipeline generates a margin of electoral return. Second, that return is gated by inherited state capacity and by the political incentives of the front-line bureaucrats who operate it, since the political value of a federal authorization shock is essentially zero in municipalities without the bureaucratic infrastructure to convert it into visible delivery and concentrates within active municipalities in those where the front-line manager turned over after the reform. National rules shape what programs look like on paper, but the local political economy of delivery is where electoral returns are produced and captured.

A natural next step is the structural counterpart to the reduced-form decomposition we report. Calibrating the augmented model with the within-subgroup first-stage moments — normalizing the active cell to $\gamma_{\text{CRAS} \times \text{turn}} = 1$ — pins the relative productivity of the no-CRAS subgroup at approximately one third of the active level and the CRAS-no-turnover subgroup at zero, and the calibrated model recovers the Frisch-Waugh-Lovell weights to within rounding. We leave a full method-of-moments estimation, with parametric inference on γ and a structural interpretation of the dose-response across subgroups, to a companion paper.

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